
Supporting Information for

Decision-Relevant Modelling of Extreme Demand for Industrial Electricity Tariffs

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APPENDIX S1. SUPPLEMENTARY METHODS, DERIVATIONS, AND ANALYSES

This Supporting Information provides technical derivations, implementation details, complete supplementary summaries, robustness analyses, and external predictive analyses referenced in the main article. The primary controlled experiment compares five predictive constructions. TAIL and EVENT-EMP use event-level histories, whereas GEV, KDE, and EMP use the corresponding monthly maxima. The primary controlled experiment contains 10,000 independent 24-month histories and nine prespecified tariff cells, giving 450,000 history–method–cell decisions. Supplementary recovery, stress, physical-clipping, and physical-misspecification experiments are separate experiments and retain their own stated analysis units.

Throughout, $M \geq 0$ denotes monthly maximum demand, F its distribution,

$$\mu_F = \mathbb{E}_F[M], \quad H_F(t) = \mathbb{E}_F[(M - t)_+].$$

The reference capacity is S , the contract level is D , the contract-threshold multiplier is α , the lower contract fraction is δ , and the exceedance-charge multiplier is κ . Optimized mode values are denoted by $V_F(m)$, the oracle value by V_F^* , oracle mode separation by γ_F , and the maximum mode-value perturbation by e_V . The retrospective stability ratio is

$$d_{MV} = \frac{2e_V}{\gamma_F}.$$

For a selected full action $\hat{a}$, regret is

$$R_F(\hat{a}) = C_F(\hat{a}) - V_F^*.$$

S1.1 Supplementary tariff identities and decision-stability details

The main article gives the tariff functionals and the principal decision-stability results. This subsection records only the atom-aware and boundary details used by the implementation.

For the contract objective

$$\phi_F(D) = D + \kappa H_F(\alpha D), \quad D \in [\delta S, S],$$

the stop-loss functional has one-sided derivatives

$$H'_{F,+}(t) = -\Pr_F(M > t), \quad H'_{F,-}(t) = -\Pr_F(M \geq t).$$

Consequently,

$$\partial \phi_F(D) = \left[1 - \kappa \alpha \Pr_F(M \geq \alpha D), 1 - \kappa \alpha \Pr_F(M > \alpha D) \right].$$

For an interior optimizer this gives

$$\Pr_F(M > \alpha D^*) \leq \frac{1}{\kappa \alpha} \leq \Pr_F(M \geq \alpha D^*).$$

Writing

$$p^* = 1 - \frac{1}{\kappa \alpha},$$

αD^* is therefore a generalized p^* -quantile. The feasible-boundary cases are handled by restricting the generalized quantile solution to

$$D \in [\delta S, S].$$

Mode stability and demand-scale radii.

For a unique oracle tariff mode m_F^* , define, for each $m \neq m_F^*$,

$$\gamma_{F,m} = V_F(m) - V_F(m_F^*),$$

and, for every $j \in \mathcal{M}$,

$$e_j = |V_{\hat{F}}(j) - V_F(j)|.$$

37 The pairwise sufficient condition is

$$38 \quad e_m + e_{m_F^*} < \gamma_{F,m} \quad \text{for every } m \neq m_F^*.$$

39 With

$$40 \quad e_V = \max_{j \in \mathcal{M}} e_j, \quad \gamma_F = \min_{m \neq m_F^*} \gamma_{F,m},$$

41 the scalar condition

$$42 \quad d_{MV} = \frac{2e_V}{\gamma_F} < 1$$

43 is a simpler conservative sufficient certificate. Both conditions use oracle quantities and are retrospective diagnostics in the
44 controlled experiment.

45 For the demand-scale stability analysis, let

$$46 \quad L_{\text{cap}} = 0, \quad L_{\text{act}} = r_d, \quad L_{\text{con}} = r_d \kappa,$$

47 and define the direct tariff-contrast constants

$$48 \quad K_{\text{cap,act}} = r_d, \quad K_{\text{cap,con}} = r_d \kappa, \quad K_{\text{act,con}} = r_d \max\{1, |1 - \kappa|\}.$$

49 For a unique oracle mode m_F^* , the corresponding analytic radii are

$$50 \quad \rho_F^{\text{contrast}} = \min_{m \neq m_F^*} \frac{\gamma_{F,m}}{K_{m,m_F^*}},$$

$$51 \quad \rho_F^{\text{pair}} = \min_{m \neq m_F^*} \frac{\gamma_{F,m}}{L_m + L_{m_F^*}}, \quad \rho_F = \frac{\gamma_F}{2L}, \quad L = r_d \max\{1, \kappa\}.$$

53 They satisfy

$$54 \quad \rho_F^{\text{contrast}} \geq \rho_F^{\text{pair}} \geq \rho_F.$$

55 Table S13 reports these oracle-dependent demand-scale radii for the nine primary tariff cells. They are analytic quantities and
56 contain neither a numerical integration allowance nor a solver allowance.

57 *Full-action loss.*

58 The exact regret decomposition separates mode-selection loss from within-mode excess. The latter is identically zero for actual
59 demand and capacity. Only contract demand can have nonzero within-mode excess. If contract demand is both oracle optimal
60 and selected, and $\hat{D}$ exactly minimizes the fitted contract objective, then

$$61 \quad R_F(a_{\text{con}}(\hat{D})) \leq 2r_d \kappa \Delta_H(F, \hat{F}),$$

62 where

$$63 \quad \Delta_H(F, \hat{F}) = \sup_{D \in \mathcal{D}_S} |H_{\hat{F}}(\alpha D) - H_F(\alpha D)|.$$

64 As an implementation check, the exact regret decomposition was verified on all 450,000 primary decision rows, with no
65 failures; the maximum absolute residual was 7.45×10^{-9} CNY/month.

66 S1.2 Analytic oracle and deterministic numerical evaluation

67 The controlled data-generating mechanism admits an analytic monthly-maximum distribution. In each month, 200 bounded
68 bulk events are generated as

$$69 \quad X^{(b)} = 45,000 + 35,000 B, \quad B \sim \text{Beta}(8, 2),$$

70 and an independent number

$$71 \quad N \sim \text{Poisson}(30)$$

72 of tail events is generated as

$$73 \quad X^{(t)} = 80,000 + Y, \quad Y \sim \text{GPD}(\xi = 0.15, \sigma = 12,000).$$

74 The monthly maximum M is the maximum over the bulk and tail events.

75 Let

$$76 \quad L = 45,000, \quad u_0 = 80,000, \quad \lambda = 30,$$

77 and let F_B denote the Beta(8, 2) distribution function. The exact monthly-maximum CDF is

$$78 \quad F_M(x) = \begin{cases} 0, & x < L, \\ \exp(-\lambda) \left[F_B\left(\frac{x-L}{u_0-L}\right) \right]^{200}, & L \leq x < u_0, \\ \exp\left[-\lambda \left\{ 1 + \xi \frac{x-u_0}{\sigma} \right\}^{-1/\xi}\right], & x \geq u_0. \end{cases}$$

79 The resulting oracle quantities are

$$80 \quad \mu_F = 148,235.371489 \text{ kW},$$

$$81 \quad q_{0.99} = 265,663.010007 \text{ kW},$$

82 and

$$83 \quad q_{0.999} = 375,513.529996 \text{ kW}.$$

84 The oracle mean and stop-loss functionals are evaluated from

$$85 \quad \mu_F = \int_0^\infty \{1 - F_M(x)\} dx$$

86 and

$$87 \quad H_F(t) = \int_t^\infty \{1 - F_M(x)\} dx.$$

88 High quantiles are obtained from the analytic distribution. The same oracle law therefore determines the reference capacity

$$89 \quad S = \frac{\mu_F}{u},$$

90 the tariff-mode values, the optimized contract level, the unique oracle mode where applicable, and the oracle mode separation γ_F .

91 Final fitted-law evaluation likewise avoids a common predictive Monte Carlo layer between $\hat{F}$ and the tariff functionals.
92 Finite discrete distributions use exact sums. The nonnegative Gaussian KDE uses analytic Gaussian-mixture expressions for
93 its mean and stop-loss function. The fitted tail-process and nonnegative-conditioned GEV constructions use deterministic one-
94 dimensional integration where required, and continuous quantiles are obtained by direct inversion or deterministic bracketed
95 root finding.

96 The deterministic numerical layer was independently checked in 55 numerical comparisons. The maximum relative
97 discrepancy was approximately

$$98 \quad 1.283 \times 10^{-11}.$$

99 No Monte Carlo approximation to the oracle is used in the reported controlled experiment.

102 S1.3 Predictive-construction implementation

103 The five primary predictive constructions are organized by their information source. TAIL and EVENT-EMP use event-level
104 observations. GEV, KDE, and EMP use the 24 observed monthly maxima. This distinction is retained because comparisons
105 across the two groups combine differences in information resolution and distributional construction, whereas TAIL versus
106 EVENT-EMP and comparisons within GEV/KDE/EMP provide same-broad-information contrasts.

107 Event-level constructions

108 *TAIL.*

109 TAIL receives event-level maxima with month identifiers. Specification selection uses the first 19 months for candidate fitting
110 and the final five months for internal predictive validation. Candidate thresholds are the empirical event-level quantiles

111 $0.90, \quad 0.925, \quad 0.95, \quad 0.975.$

112 For a candidate numerical threshold u , exceedances are strict: only observations $Z > u$ enter the excess sample

113
$$Y = Z - u.$$

114 The excess law is fitted by GPD maximum likelihood with location fixed at zero.

115 Monthly exceedance counts are formed over all fitting months, including months with zero exceedances. If $\bar{N}$ and s_N^2 denote
116 the empirical monthly count mean and variance, the count law is Poisson when the variance-to-mean ratio is at most 1.2 or
117 when the empirical mean is nonpositive. Otherwise the negative-binomial method-of-moments parameters are

118
$$r = \frac{\bar{N}^2}{s_N^2 - \bar{N}}, \quad p = \frac{r}{r + \bar{N}}.$$

119 The corresponding probability-generating functions are

120
$$P_N(z) = \exp\{\lambda(z - 1)\}$$

121 for the Poisson case and

122
$$P_N(z) = \left[\frac{p}{1 - (1 - p)z} \right]^r$$

123 for the negative-binomial case.

124 Candidate screening requires at least 100 strict exceedances. Shape stability is assessed from 50 nonparametric excess-
125 sample bootstrap fits, with at least 40 successful fits required and the resulting 95% shape-interval width not exceeding 0.8. The
126 success requirement is 80% of the configured replicate count, with at least 40 fits in the primary experiment. The probability-
127 transformed fitted tail must satisfy a Kolmogorov–Smirnov p -value threshold of 0.01. For a negative-shape candidate, the
128 natural finite endpoint

129
$$e_u = u - \frac{\hat{\sigma}_u}{\hat{\xi}}$$

130 must satisfy

131
$$e_u \geq \max_i X_i - 10^{-9},$$

132 where X_i are the fitting-period event maxima in kW. Candidate fits must have a finite validation score and satisfy

133
$$\hat{\xi} < 1,$$

134 which ensures a finite first moment.

135 For each fitted candidate, we simulate $m = 2,000$ monthly maxima $Z_1, \dots, Z_m$ on the fitted candidate's natural support,
136 without artificial upper clipping, and estimate their density with a Gaussian kernel:

137
$$\hat{f}_m(y) = \frac{1}{mh} \sum_{j=1}^m \phi\left(\frac{y - Z_j}{h}\right), \quad h = s_Z m^{-1/5},$$

138 where ϕ is the standard normal density and s_Z is the sample standard deviation with denominator $m-1$. This is Scott's bandwidth
139 as implemented by `scipy.stats.gaussian_kde`. For the five held-out monthly maxima $Y_1, \dots, Y_5$, the score is

140
$$\ell_{\text{val}} = \frac{1}{5} \sum_{t=1}^5 \log \left\{ \max \left(\hat{f}_m(Y_t), \epsilon_{\text{tiny}} \right) \right\}, \quad \epsilon_{\text{tiny}} = 2.2250738585072014 \times 10^{-308}.$$

141 A numerically constant predictive sample has an undefined score and is excluded. If $\mathcal{P}$ denotes the candidates that pass the
 142 screening and finite-mean conditions, the selection score is

$$143 \quad \ell_{\text{val}} - 0.10 \left| \hat{\xi} - \text{median}_{j \in \mathcal{P}}(\hat{\xi}_j) \right|.$$

144 An exact tie is resolved toward the lower threshold quantile.

145 The 19/5 split is used only for specification selection. After the numerical threshold is selected, that threshold is held fixed
 146 and the GPD parameters, exceedance-count law, and empirical subthreshold component are reconstructed using all 24 months.

147 If no admissible finite-mean primary candidate is available but an eligible threshold has at least 50 exceedances, the procedure
 148 uses an exponential-tail fallback with

$$149 \quad \xi = 0$$

150 and scale equal to the corresponding mean excess. If no such threshold is available, the monthly-maxima empirical distribution
 151 is used. An invalid all-history GPD refit likewise invokes the prescribed fallback branch.

152 For a final primary TAIL fit, let F_0 denote the empirical CDF of the subthreshold monthly-maximum component, let G_u
 153 denote the fitted GPD excess CDF, and let P_N denote the fitted exceedance-count probability-generating function. The monthly-
 154 maximum CDF is

$$155 \quad \hat{F}_{\text{TAIL}}(x) = \begin{cases} F_0(x)P_N(0), & x < u, \\ P_N\{G_u(x-u)\}, & x \geq u. \end{cases}$$

156 The final mean is evaluated as

$$157 \quad \mu_{\hat{F}_{\text{TAIL}}} = H_{\hat{F}_{\text{TAIL}}}(0),$$

158 the stop-loss function is obtained by deterministic integration of the fitted survival function, and continuous quantiles are
 159 obtained by deterministic inversion. Thus the simulation used for internal candidate validation is not reused as the final
 160 representation of the tariff-relevant functionals.

161 *EVENT-EMP.*

162 EVENT-EMP uses the same event-level history as TAIL but imposes no threshold or parametric tail/count model. Let $\hat{G}_E$
 163 denote the pooled empirical CDF of all event-level maxima in the 24-month history, and let N_m denote the observed total event
 164 count in month m . The induced monthly-maximum CDF is

$$165 \quad \hat{F}_{\text{EE}}(x) = \frac{1}{24} \sum_{m=1}^{24} \hat{G}_E(x)^{N_m}.$$

166 This is a finite discrete distribution. Its probability masses are the jumps of $\hat{F}_{\text{EE}}$; its mean and stop-loss function are evaluated
 167 exactly by finite sums, and its quantiles use the discrete left inverse. There is no dedicated fallback branch.

168 TAIL and EVENT-EMP therefore use the same broad event-level information source but different distributional constructions.
 169 TAIL adds threshold selection, a parametric excess law and a fitted exceedance-count law, whereas EVENT-EMP combines a
 170 pooled empirical event distribution with the empirical distribution of observed monthly event counts.

171 Monthly-maxima constructions

172 *GEV.*

173 The controlled-experiment GEV construction uses the 24 observed monthly maxima. It is fitted using L-moments/probability-
 174 weighted moments rather than maximum likelihood. L-skewness is inverted deterministically for the EVT shape parameter
 175 within

$$176 \quad [-0.90, 0.90],$$

177 and the location and scale parameters follow from the associated L-moment equations.

178 A successful raw GEV fit is conditioned on the nonnegative part of its model-implied natural support. If more than 1% of the
 179 raw fitted GEV probability lies below zero, or if the L-moment fit or resulting parameters are invalid, the procedure uses the

empirical monthly-maxima distribution. Successful parametric fits satisfy the finite-mean requirement. When the fitted EVT shape is negative, its model-implied finite upper endpoint is retained; otherwise the upper support is unbounded.

The conditioned GEV mean and stop-loss function are evaluated deterministically over this support, using one-dimensional integration where required. Quantiles use the conditioned inverse distribution. This GEV construction is distinct from the K-GEV procedure used in the external manufacturing-load analysis.

KDE.

KDE is fitted to the 24 monthly maxima $M_1, \dots, M_n$, with $n = 24$, using bandwidth

$$h = 1.06 s n^{-1/5},$$

where s is the sample standard deviation. The Gaussian mixture is conditioned on nonnegative demand. If

$$\tilde{F}(x) = \frac{1}{n} \sum_{i=1}^n \Phi\left(\frac{x - M_i}{h}\right)$$

denotes the unconditioned Gaussian-mixture CDF and

$$Z_+ = 1 - \tilde{F}(0),$$

then

$$\hat{F}_{\text{KDE}}(x) = \begin{cases} 0, & x < 0, \\ \frac{\tilde{F}(x) - \tilde{F}(0)}{Z_+}, & x \geq 0. \end{cases}$$

For $t \geq 0$, its stop-loss function is

$$H_{\hat{F}_{\text{KDE}}}(t) = \frac{1}{nZ_+} \sum_{i=1}^n \left[(M_i - t) \Phi\left(\frac{M_i - t}{h}\right) + h \phi\left(\frac{t - M_i}{h}\right) \right],$$

where Φ and ϕ are the standard normal CDF and PDF. Setting $t = 0$ gives the mean. Quantiles are obtained by deterministic inversion of the conditioned mixture CDF. If fewer than two observations are available or the observed variability is numerically degenerate, the implementation uses the empirical distribution.

EMP.

EMP assigns empirical probability to the 24 observed monthly maxima, with duplicate support values aggregated. For support points x_j with empirical probabilities p_j ,

$$\mu_{\hat{F}_{\text{EMP}}} = \sum_j p_j x_j,$$

and

$$H_{\hat{F}_{\text{EMP}}}(t) = \sum_j p_j (x_j - t)_+.$$

Quantiles use the empirical left inverse. No predictive-bootstrap layer is introduced into the final tariff-functional evaluation.

S1.4 From predictive laws to tariff actions

Every fitted predictive law is passed through the same tariff-action map. For the evaluated policies, $\kappa\alpha > 1$, and the interior contract probability is

$$p^* = 1 - \frac{1}{\kappa\alpha}.$$

For a fitted law $\hat{F}$, define the left generalized quantile

$$\hat{q}_p = \inf\{x : \hat{F}(x) \geq p\}.$$

212 The final contract level is

$$213 \quad \hat{D} = \text{clip} \left(\frac{\hat{q}_{p^*}}{\alpha}, \delta S, S \right).$$

214 Thus lower and upper boundary solutions are handled directly through the feasible interval. No predictive Monte Carlo
215 candidate search or local order-statistic candidate set is used.

216 The fitted optimized tariff-mode values are

$$217 \quad V_{\hat{F}}(\text{cap}) = r_c S,$$

$$218 \quad V_{\hat{F}}(\text{act}) = r_d \mu_{\hat{F}},$$

220 and

$$221 \quad V_{\hat{F}}(\text{con}) = r_d \left\{ \hat{D} + \kappa H_{\hat{F}}(\alpha \hat{D}) \right\}.$$

222 The selected tariff mode minimizes these deterministic mode values. Exact mode-value ties are resolved by the fixed
223 lexicographic order

$$224 \quad \text{actual demand} \prec \text{capacity} \prec \text{contract demand}.$$

225 For discrete predictive laws, the left-quantile convention fixes the generalized-quantile representative. For continuous laws,
226 quantiles are obtained directly or by deterministic root inversion.

227 Although

$$228 \quad V_F(\text{cap}) = r_c S$$

229 does not itself contain a distributional functional, selection of the capacity mode depends on its comparison with the esti-
230 mated actual-demand and contract-demand mode values. Under contract demand, the complete action additionally contains D ;
231 consequently, tariff-mode correctness and full-action optimality are separate endpoints.

232 S1.5 Primary comparisons and high- p^* tariff stress

233 The complete nine-cell primary summaries, mode-versus-within-contract regret decomposition, and core paired contrasts are
234 reported in the main article. Supplementary Table S12 gives all ten region-specific paired comparisons for the final 10,000-
235 history primary run. It is a complete reference table; the ten contrasts are not treated as one confirmatory family.

236 The high- p^* stress analysis evaluates three target quantiles over ten utilization cells, using 10,000 histories and five predictive
237 methods. Figure S1 and Table S1 report the history-level regret distributions, exact-zero-regret rates, and same-information
238 paired comparisons. This stress design is separate from the nine-cell primary analysis.

239 S1.6 Functional-error associations and certificate-conditioned loss

240 Figure S2 summarizes the within-method association of regret with absolute mean, stop-loss, and $q_{0.99}$ relative errors, and the
241 concentration of regret among the largest-loss decisions. Table S7 reports the paired functional-correlation contrasts with simul-
242 taneous max- $|T|$ intervals. Table S2 reports within-contract loss among decisions stratified by certificate status. The certificate
243 is a sufficient condition; observations outside the certified set are not classified as violations.

244 S1.7 Numerical bounds and fitting-support sensitivity

245 The theoretical sufficient conditions in the main article are summarized separately from their numerical implementation. The
246 numerical audit uses upper constructions for the relevant distributional perturbations and a common numerical full-action solver
247 allowance. For the primitive-action perturbation, the reported numerical upper quantity is

$$248 \quad \varepsilon_{\text{num}} = \max \left\{ r_d \left| \hat{\mu} - \mu_F \right|, r_d \kappa \Delta_{H, \text{upper}} \right\}.$$

249 It is a numerical upper bound on primitive-action perturbation, not the mode-value error e_V and not an exact analytic sup-
 250 remum. The feasible-threshold stop-loss quantity uses a direct 513-node estimate and a numerical upper allowance based on the
 251 monotone-CDF envelope and half-grid spacing; it is a numerical upper bound on the feasible stop-loss perturbation.

252 The primitive numerical sufficient-condition check is

$$253 \quad 2\varepsilon_{\text{num}} + \zeta_{\text{full,num}} < \gamma_F.$$

254 For each non-oracle competitor m , the direct contrast-Wasserstein check is

$$255 \quad K_{m,m_F^*} W_{1,\text{upper}} + \zeta_{\text{full,num}} < \gamma_{F,m},$$

256 where $K_{\text{cap,act}} = r_d$, $K_{\text{cap,con}} = r_d \kappa$, and $K_{\text{act,con}} = r_d \max\{1, |1 - \kappa|\}$. The mode-wise pairwise Wasserstein comparator instead
 257 uses

$$258 \quad (L_m + L_{m_F^*}) W_{1,\text{upper}} + \zeta_{\text{full,num}} < \gamma_{F,m},$$

259 for every non-oracle competitor, and the uniform Wasserstein check is

$$260 \quad 2LW_{1,\text{upper}} + \zeta_{\text{full,num}} < \gamma_F, \quad L = r_d \max\{1, \kappa\}.$$

261 The numerical Wasserstein upper quantity is based on CDF bounds; the point estimates reported in Table S6 use numerical
 262 quadrature. The solver allowance and distributional numerical quantities are floating-point implementation bounds without
 263 directed-rounding or interval-arithmetic guarantees. Table S14 reports the observed implementation-audit results: the primitive-
 264 perturbation check was satisfied by 26,975/450,000 (5.994444%); the direct contrast-Wasserstein check by 57,501/450,000
 265 (12.778000%); the mode-wise pairwise-Wasserstein check by 26,393/450,000 (5.865111%); and the uniform-Wasserstein
 266 check by 5,329/450,000 (1.184222%). No observed mode-selection violations occurred among rows satisfying any of these
 267 four numerical checks.

268 *Fitting-support sensitivity.*

269 The primary controlled analysis evaluates each fitted law on the estimator-specific support described in Section S1.3. To assess
 270 sensitivity to finite fitting bounds, a paired analysis uses the first 1,000 primary histories and the same nine tariff cells. Three
 271 training-derived upper bounds are considered:

$$272 \quad B_{\text{tr}}^{(10)} = 10 \max_{1 \leq m \leq 24} M_m, \quad B_{\text{tr}}^{(20)} = 20 \max_{1 \leq m \leq 24} M_m, \quad B_{\text{tr}}^{(50)} = 50 \max_{1 \leq m \leq 24} M_m,$$

273 where $M_1, \dots, M_{24}$ are the observed training monthly maxima. Because the data-generating distribution is known in the
 274 controlled experiment, the sensitivity analysis also includes the oracle-referenced diagnostic bound

$$275 \quad B_{\text{or}} = 10q_{0.999}.$$

276 This diagnostic specification is used only in the sensitivity analysis. The same histories and tariff cells are retained under every
 277 support rule. The support treatment affects the TAIL and GEV constructions; EVENT-EMP, KDE, and EMP are unchanged by
 278 construction.

279 Panels (d–f) of Figure S3 summarize the paired support-policy comparisons relative to the primary specification. Panels (d)
 280 and (e) show cellwise paired mean changes in regret for TAIL and GEV, respectively, with one paired Monte Carlo standard
 281 error, and panel (f) shows the percentage of TAIL tariff-mode decisions that change. Table S11 reports, for both TAIL and GEV,
 282 the paired mean regret change, its paired standard error, and the percentage of changed tariff-mode decisions in each tariff
 283 cell. Across the displayed support rules and cells, the changed-mode percentage does not exceed 0.20% for TAIL or 0.40% for
 284 GEV. The largest absolute cellwise paired mean-regret changes are 0.062 and 0.726 thousand CNY/month for TAIL and GEV,
 285 respectively.

286 S1.8 Parameter recovery and threshold-fixed bootstrap interval coverage

287 The parameter-recovery experiment is separate from the 10,000-history primary comparison. It uses 200 histories at each of 12,
288 24, and 60 months under a Poisson–GPD mechanism with generating threshold 80,000 kW,

$$289 \quad \xi = 0.15, \quad \sigma = 12,000 \text{ kW},$$

290 and a Poisson rate of 30 tail events per month.

291 The threshold continues to be selected by the fitted procedure rather than fixed at the generating threshold. Scale recovery is
292 therefore evaluated relative to the threshold-specific scale implied by GPD threshold stability.

293 The mean absolute errors of the fitted shape parameter are

$$294 \quad 0.0699, \quad 0.0578, \quad 0.0343$$

295 for 12, 24, and 60 months, respectively. Figure S4 reports the complete shape-error, relative-scale-error, and relative- $q_{0.99}$ error
296 distributions.

297 For each history of length n months, we construct nominal 95% nonparametric percentile intervals for the GPD shape and
298 scale. The selected threshold $\hat{v}$ is held fixed. We draw 100 bootstrap samples with replacement from the excesses in the first
299 $\lfloor 0.8n \rfloor$ months, preserving the excess count, and fit a GPD with location fixed at zero to each sample. The interval endpoints
300 are the 2.5th and 97.5th percentiles of the valid parameter estimates; at least 75 finite fits with positive scale are required. Shape
301 coverage is evaluated against $\xi = 0.15$, and scale coverage against $\sigma(\hat{v}) = 12000 + 0.15(\hat{v} - 80000)$. Table S10 reports coverage
302 across 200 histories and 95% Clopper–Pearson intervals for its Monte Carlo uncertainty. Thresholds vary across histories and
303 remain fixed within each bootstrap calculation.

304 S1.9 Physical clipping identity and sensitivity

305 Let latent event-level demand values be X_i , and let physical clipping operate at level c . Clipping commutes with maximization:

$$306 \quad \min \left(\max_i X_i, c \right) = \max_i \min(X_i, c).$$

307 Thus clipping individual events before monthly maximization yields the same clipped monthly maximum as clipping the
308 corresponding original monthly maximum.

309 Let

$$310 \quad M_c = \min(M, c)$$

311 and let F_c denote its distribution. Pointwise,

$$312 \quad M_c = M - (M - c)_+,$$

313 so

$$314 \quad \mathbb{E}[M_c] = \mathbb{E}[M] - H_F(c).$$

315 For every $t \leq c$,

$$316 \quad (\min(M, c) - t)_+ = (M - t)_+ - (M - c)_+,$$

317 and therefore

$$318 \quad H_{F_c}(t) = H_F(t) - H_F(c), \quad t \leq c.$$

319 For $t \geq c$,

$$320 \quad H_{F_c}(t) = 0.$$

321 The equality case $t = c$ is included in the first identity.

322 The clipping experiment uses four physical regimes:

$$\begin{aligned}
 &\text{UNCAPPED : } c = 10q_{0.999}, \\
 &\text{DISTANT : } c = \frac{q_{0.999}}{0.85}, \\
 &\text{Q999 : } c = q_{0.999}, \\
 &\text{BINDING : } c = \frac{q_{0.999}}{1.25}.
 \end{aligned}$$

324 “UNCAPPED” is an experimental label for a distant finite physical clipping level. Its value $c = 10q_{0.999}$ defines a clipping
 325 regime only in this robustness experiment. The clipping level c is applied to generated demand and is distinct from the fitted-law
 326 support treatment used in the primary controlled analysis.

327 The experiment uses 12- and 24-month histories,

$$328 \quad u \in \{0.62, 0.63, 0.64\},$$

329 and 200 histories for each history-length–clipping–utilization setting. TAIL, KDE, EMP, and the robustness-only GATE selector
 330 are retained. In this experiment, normalized regret is one-month regret divided by one monthly capacity fee; it is distinct from
 331 the primary normalized regret R_F/V_F^* .

332 GATE selects TAIL when its fit status is `PASS`, the fraction of reconstructed maxima exceeding the support bound is at most
 333 0.01, and two consecutive available threshold candidates pass screening with shape estimates of the same sign. Candidates
 334 are ordered by their threshold quantiles 0.90, 0.925, 0.95, 0.975; the qualifying pair need not include the threshold ultimately
 335 selected by TAIL. Each member must have at least 100 exceedances, a nominal 95% bootstrap shape-interval width at most 0.8,
 336 a uniform-transform KS p -value at least 0.01, and a compatible endpoint when its shape is negative, as defined in S1.3. GATE
 337 otherwise selects KDE. Shape-width screening uses 50 bootstrap resamples here and 500 in the physical-pipeline experiment
 338 in S1.10, with at least 80% valid refits. The support bound is the clipping level c here and 120 MW in S1.10. Two zero shape
 339 estimates count as having the same sign.

340 Figure S5 reports paired TAIL-minus-baseline ECDFs after averaging the three utilization cells within history. Negative
 341 values favour TAIL. ECDFs are used because the interquartile ranges of all 16 plotted contrast groups are zero and boxplots
 342 would obscure their nonzero tails.

343 S1.10 Physical-pipeline stress at $\alpha = 1.20$

344 The physical-pipeline experiment evaluates high-amplitude, high-frequency, and clustered-event mechanisms at $\alpha = 1.20$. Each
 345 method, training length, and tariff cell is evaluated on 200 paired histories, for 72,800 decision records. Figure S6 and Table S3
 346 report the distributional and cell-aggregated results.

347 S1.11 Joint negative-binomial–Weibull challenge

348 The joint misspecification challenge combines a negative-binomial event count with Weibull exceedances. Figure S7 shows
 349 the cellwise distribution of regret across the actual, contract, and capacity oracle regions. Table S4 gives the complete cell-
 350 by-method results, and Table S5 gives the full regional paired contrasts. This experiment assesses a joint count and severity
 351 misspecification and is not used as a replacement for the physical-pipeline experiment above.

352 S1.12 Crossed tariff geometry without an artificial upper fitting bound

353 The crossed-grid analysis evaluates five methods over 19 utilization values at $\alpha \in \{1.05, 1.10, 1.15, 1.20\}$ using 1,000 shared
 354 histories, for 380,000 decisions. Figure S8 shows mean normalized full-action regret and oracle-mode accuracy. The curves
 355 connect discrete evaluated tariff cells for visual comparison; no interpolation or invariant within-region ranking is implied.

training month, pools these complete-block maxima over the training months, and resamples them with replacement. Only blocks containing 96 observations at consecutive 15-minute timestamps are eligible. For a target month containing d days, the procedure samples d eligible 24-hour block maxima with replacement and takes their maximum as one synthetic monthly maximum. The external predictive scoring layer uses 50,000 such draws. Thus the implemented procedure resamples complete 24-hour block maxima rather than raw 15-minute observations. Across all 32 factory–origin fits, the 402,570 eligible blocks had consecutive 15-minute timestamps. These verified block sets were used for K-BLOCK prediction.

400 *K-POINT*.

401 K-POINT is a degenerate predictive reference concentrated at the mean of the normalized training monthly maxima.

402 *K-GTAIL*.

403 K-GTAIL is a runs-declustered peaks-over-threshold construction fitted to the normalized fixed-15 training series. Candidate threshold quantiles are

$$405 \quad \{0.90, 0.925, 0.95, 0.975\},$$

406 and candidate run lengths are

$$407 \quad \{30, 60, 120\} \text{ minutes.}$$

408 Exceedances are strict. Exceedances separated by no more than the candidate run length belong to the same run cluster, and each cluster is represented by its maximum excess. GPD excesses are fitted by maximum likelihood with location fixed at zero.

410 Every threshold–run candidate is evaluated through the leave-one-training-month-out procedure above. Within a fold, candidate predictive performance is measured on the held-out training-month maximum, and fold scores are averaged before specification selection. The selected threshold–run specification is then refitted using all training months.

413 The final external tail reconstruction uses a Poisson cluster-count law with rate estimated from the training clusters. A bootstrapped subthreshold monthly component is combined with simulated Poisson cluster counts and GPD excess severities to form predictive monthly maxima. The final external predictive representation contains 50,000 monthly-maximum draws.

416 K-GTAIL also retains prespecified fit screening and fallback branches as part of the complete finite-sample procedure. If an admissible gated tail specification is unavailable, or if the selected all-training fit fails the required diagnostic checks, the procedure can use an exponential-tail fallback with $\xi = 0$ or an empirical monthly-maxima fallback. These fallback predictions remain in the formal score evaluation rather than being removed conditionally on fit status.

420 After leave-one-training-month-out selection, the all-training K-GTAIL refit applies the final finite-sample gates directly. It requires at least 50 run clusters, a 95% parametric-bootstrap interval for the GPD shape with width no greater than 0.8, a parametric-bootstrap Kolmogorov–Smirnov (p)-value of at least 0.05, and compatibility of any negative-shape endpoint with the observed final peaks and the normalized support bound 10. The shape and KS quantities use 500 parametric bootstrap replicates; at least 80% successful refits are required for these quantities to be considered valid. Candidate selection is by the mean leave-one-training-month-out validation log score, with ties broken by the lower threshold quantile and then the shorter run length. Each fold uses 2,000 predictive draws and the Gaussian-kernel log score defined in S1.3, evaluated at its held-out monthly maximum. Both the draws and the observation are divided by the mean monthly maximum of that fold’s fitting months. If the predictive draws have numerically zero standard deviation, the scoring density is Gaussian with mean $\bar{Z}$ and standard deviation $\max(10^{-6}|\bar{Z}|, 10^{-8})$; the same density floor ϵ_{tiny} applies. For a negative-shape final fit, the endpoint gate is $\max_i X_i \leq u - \hat{\sigma}_u/\hat{\xi} \leq 10$, with X_i the normalized final cluster peaks.

431 If the final gate fails, the implementation compares exponential-tail and empirical fallbacks using leave-one-training-month-out scores. The exponential fallback is used only when its score is at least as good as the empirical fallback and the final fit has at least 25 clusters; otherwise the empirical monthly-maxima fallback is used. A final fit with fewer than three excesses also uses the empirical fallback.

435 *K-UGPD*.

436 K-UGPD is an auxiliary ungated POT/GPD construction using the same high-frequency training source and candidate threshold/run framework. It retains the tail construction without the complete K-GTAIL screening/fallback layer and is reported only as an external supplementary comparator.

439 *K-GEV.*

440 K-GEV is the external block-maxima parametric comparator. It fits a GEV distribution by maximum likelihood using SciPy's
 441 generalized-extreme- value implementation on the normalized training monthly maxima. At least six nondegenerate training
 442 maxima are required; the fitted shape, location, and scale must be finite and the scale must be positive. The fit is represented by
 443 50,000 predictive draws. A draw outside the nonnegative interval $([0,10])$ is counted in the truncation fraction; if this fraction
 444 exceeds 1%, if a nonfinite draw occurs, or if fitting fails, the procedure falls back to the empirical distribution of the normalized
 445 training monthly maxima. Otherwise the draws are clipped to $([0,10])$. This external construction is distinct from the controlled
 446 L-moment/PWM GEV implementation; no controlled-experiment estimation algorithm is transferred to K-GEV.

447 *Predictive scores.*

448 Whole-distribution predictive performance is evaluated by the continuous ranked probability score³,

$$449 \quad \text{CRPS}(F, y) = \mathbb{E}|X - y| - \frac{1}{2} \mathbb{E}|X - X'|,$$

450 where X and X' are independent under F .

451 For quantile level τ , pinball loss is

$$452 \quad \rho_\tau(y - q_\tau) = (y - q_\tau) [\tau - \mathbf{1}\{y - q_\tau < 0\}].$$

453 The formal quantile-specific endpoints are the losses at

$$454 \quad \tau = 0.95 \quad \text{and} \quad \tau = 0.99.$$

455 The upper-tail score $\text{qwCRPS}_{0.90}$ is the factor-two trapezoidal integration of pinball loss over the prespecified upper-tail
 456 quantile grid

$$457 \quad \mathcal{Q} = \{0.900, 0.905, \dots, 0.995\}.$$

458 Writing $q_j = \hat{F}^{-1}(\tau_j)$ and $\rho_j = \rho_{\tau_j}(y - q_j)$ for the predictive quantile and pinball loss at $\tau_j \in \mathcal{Q}$, the implemented discrete score is

$$459 \quad \text{qwCRPS}_{0.90} = 2 \sum_{j=1}^{|\mathcal{Q}|-1} \frac{\tau_{j+1} - \tau_j}{2} (\rho_j + \rho_{j+1}).$$

460 The score is evaluated on this fixed grid and is not reinterpreted as an interchangeable full-distribution CRPS.

461 For each method b and quantile level τ , empirical exceedance frequency is summarized by

$$462 \quad \hat{e}_{b,\tau} = \frac{1}{32} \sum_{f=1}^8 \sum_{o=1}^4 \mathbf{1}\{y_{f,o} > q_{b,f,o,\tau}\}.$$

463 The nominal exceedance frequency is $1 - \tau$. The mean squared- exceedance statistic satisfies

$$464 \quad \bar{B}_{b,\tau} = (1 - \tau)^2 + (2\tau - 1)\hat{e}_{b,\tau}.$$

465 It is therefore an affine summary of empirical exceedance frequency at method-specific thresholds. Calibration is described
 466 relative to the nominal frequency, while comparative predictive loss is evaluated using CRPS, upper-quantile-weighted pinball
 467 loss, and the two individual pinball losses.

468 For aggregate method comparisons, the four origin-specific scores are first averaged within eligible factory, and differences
 469 are paired across the eight factories. Table S9 reports paired mean differences with pointwise 95% paired t intervals using seven
 470 degrees of freedom. The external analysis evaluates predictive distributions; no observed tariff action or realized electricity bill
 471 is available in this dataset.

472 **S1.14 External predictive scores and fitted-branch summaries**

473 The external analysis uses eight eligible factories and four rolling origins per factory. Table S8 reports formal fit-status counts
 474 across the 32 factory–origin fits for each method. Fallback fits remain part of the evaluated procedures. Table S9 reports paired

predictive-score contrasts between K-GTAIL and the listed baselines. The score-specific intervals use eight factory-level paired differences; score contrasts are descriptive and do not imply a single winner across predictive targets. No real tariff bills are observed in this dataset.

S1.15 Numerical solver allowance

For a fitted distribution $\hat{F}$, the contract-mode objective is

$$C_{\hat{F},\text{con}}(D) = r_d \left\{ D + \kappa H_{\hat{F}}(\alpha D) \right\}, \quad D \in \mathcal{D}_S = [\delta S, S].$$

At a numerical contract level $\tilde{D}$, its one-sided subgradients are

$$g_-(\tilde{D}) = r_d \left\{ 1 - \kappa \alpha \Pr_{\hat{F}}(M \geq \alpha \tilde{D}) \right\}, \quad g_+(\tilde{D}) = r_d \left\{ 1 - \kappa \alpha \Pr_{\hat{F}}(M > \alpha \tilde{D}) \right\}.$$

A valid subgradient $s \in [g_-(\tilde{D}), g_+(\tilde{D})]$ is chosen nearest to zero. Convexity gives, for every feasible D ,

$$C_{\hat{F},\text{con}}(D) \geq C_{\hat{F},\text{con}}(\tilde{D}) + s(D - \tilde{D}).$$

Therefore the fitted-objective gap is bounded by

$$B_s(\tilde{D}) = \max_{D \in [\delta S, S]} s(\tilde{D} - D) = \begin{cases} s(\tilde{D} - \delta S), & s \geq 0, \\ -s(S - \tilde{D}), & s < 0. \end{cases}$$

The numerical contract allowance adds a scale-based floating-point term,

$$\zeta_{\text{con,num}} = B_s(\tilde{D}) + 512 \epsilon_{\text{mach}} \max \left\{ 1, r_d(\tilde{D} + \kappa \alpha \tilde{D}) \right\}.$$

Actual-demand and capacity modes are singleton actions. The full-action numerical allowance therefore combines the contract gap, saved-versus-recomputed objective-evaluation allowances, and any saved candidate selection gap. Writing η_{sel} for the objective-evaluation allowance of the selected mode, $\eta_{\text{max}} = \max_{j \in \mathcal{M}} \eta_j$ for the largest mode allowance, and θ for the nonnegative saved candidate-selection gap, the common implementation allowance is

$$\zeta_{\text{full,num}} = \zeta_{\text{con,num}} + \eta_{\text{sel}} + \eta_{\text{max}} + \theta.$$

For each mode j , the numerical objective-evaluation allowance is

$$\eta_j = \left| \hat{V}_{j,\text{recomputed}} - \hat{V}_{j,\text{saved}} \right| + 512 \epsilon_{\text{mach}} \max \left\{ 1, \left| \hat{V}_{j,\text{recomputed}} \right|, \left| \hat{V}_{j,\text{saved}} \right| \right\}.$$

The saved candidate-selection gap is

$$\theta = \max \left\{ 0, \hat{V}_{\tilde{m},\text{saved}} - \min_{j \in \mathcal{M}} \hat{V}_{j,\text{saved}} \right\}.$$

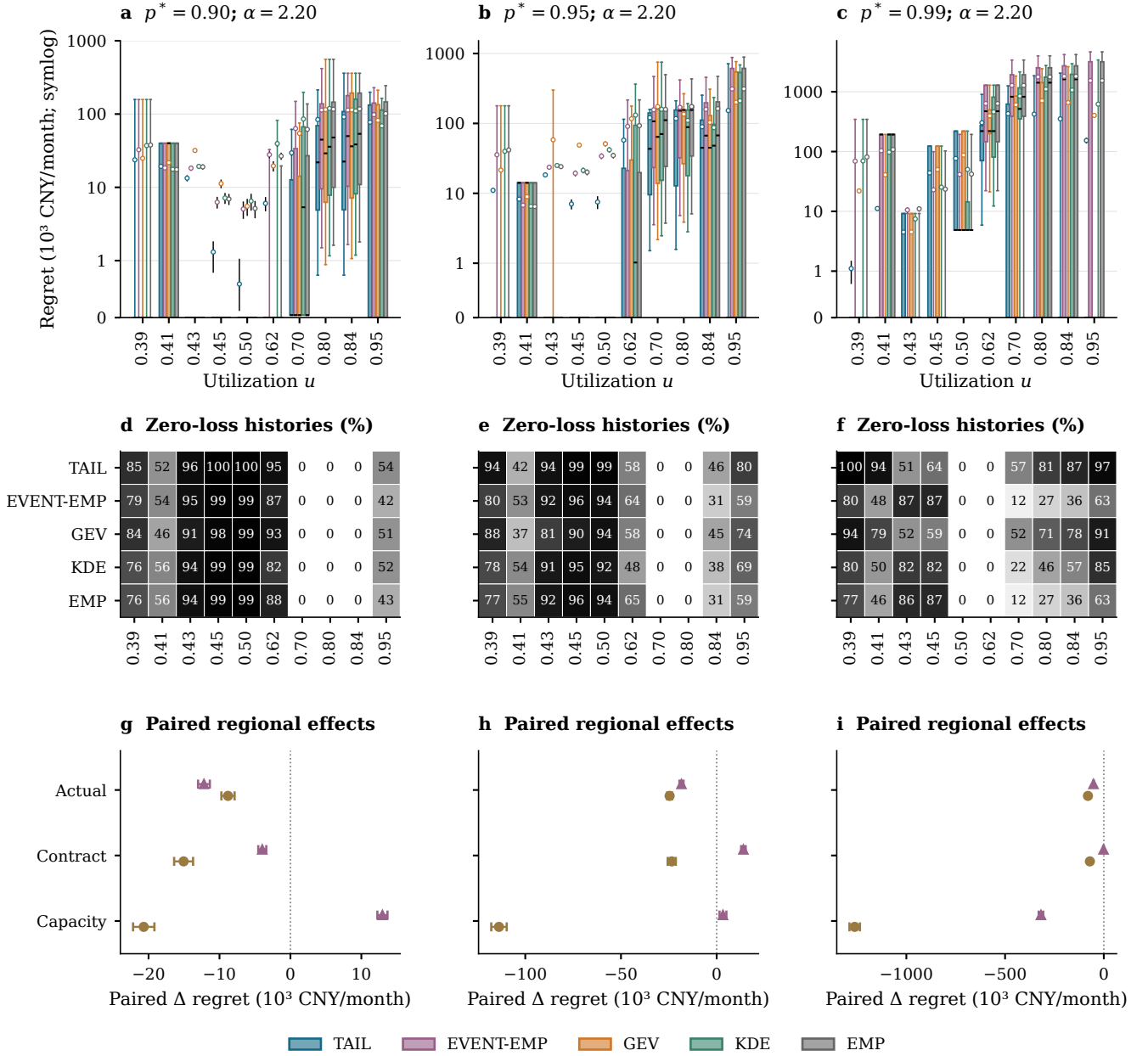
These terms use saved-versus-recomputed objective differences with a scale-based round-off term; they are numerical allowances, not proven floating-point error bounds. A 307-case implementation cross-check compared contract-objective excess with the fitted quantile-reference objective; all 307 cases satisfied the subgradient numerical bound up to the stated 10^{-3} CNY comparison tolerance.

The mathematical subgradient inequality and the reduction from a valid contract objective-gap bound to a full-action allowance are exact. The floating-point evaluations used in the numerical implementation do not use directed rounding or interval arithmetic. Accordingly, $\zeta_{\text{full,num}}$ is treated as a numerical solver allowance rather than an interval-certified error bound.

S1.16 Supplementary source and table index

The supplementary figure and table ownership is as follows.

-
- 507 • Figures S1–S8: high- p^* tariff stress; functional-error and certificate diagnostics; numerical distribution-to-decision bounds
 - 508 and fitting-support sensitivity; parameter recovery; physical clipping; physical-pipeline stress; joint NB–Weibull challenge;
 - 509 and crossed tariff geometry, respectively.
 - 510 • Table S1: high- p^* paired comparisons;
 - 511 • Table S2: certificate-conditioned within-contract loss;
 - 512 • Table S3: physical-pipeline stress;
 - 513 • Tables S4–S5: joint NB–Weibull challenge;
 - 514 • Table S6: numerical Wasserstein quantities;
 - 515 • Table S7: functional-error contrasts;
 - 516 • Table S8: external fit-status counts;
 - 517 • Table S9: external predictive-score contrasts;
 - 518 • Table S10: threshold-fixed bootstrap coverage;
 - 519 • Table S11: fitting-support sensitivity;
 - 520 • Table S12: primary regional paired contrasts;
 - 521 • Table S13: primary tariff geometry and analytic demand-scale radii;
 - 522 • Table S14: numerical sufficient-condition audit.



Boxes: P25–P75; whiskers: P10–P90; median tick; mean circle. 10,000 histories per method and cell.

Contrasts: brown circle TAIL – EVENT-EMP; purple triangle GEV – KDE. Mean and paired t95 CI.

FIGURE S1 High-contract-quantile stress experiment at $\alpha = 2.20$, comprising 10,000 histories, five methods, three target quantiles, and ten utilization cells per target quantile (1,500,000 decisions). Columns correspond to target quantiles 0.90, 0.95, and 0.99. (a–c) Cellwise regret distributions: boxes span P25–P75, whiskers P10–P90, black ticks denote medians, and open circles denote means; black intervals at means are pointwise 95% Student t intervals (9,999 degrees of freedom). The symlog ordinate retains zero, is linear below one thousand CNY per month, and logarithmic above it; ranges vary by column. (d–f) Percentages of exactly zero regret, with the same grayscale mapping from 0 to 100 in all panels. Printed values are rounded to whole percentages; a printed 100 need not imply every history has zero regret. (g–i) Same-information paired regional mean differences: TAIL minus EVENT-EMP (circles) and GEV minus KDE (triangles). Cells are equally weighted within history and oracle region before differences are formed; intervals are paired t95 intervals. Negative values favour the first method. Data were reconstructed from saved fits without refitting or new simulation.

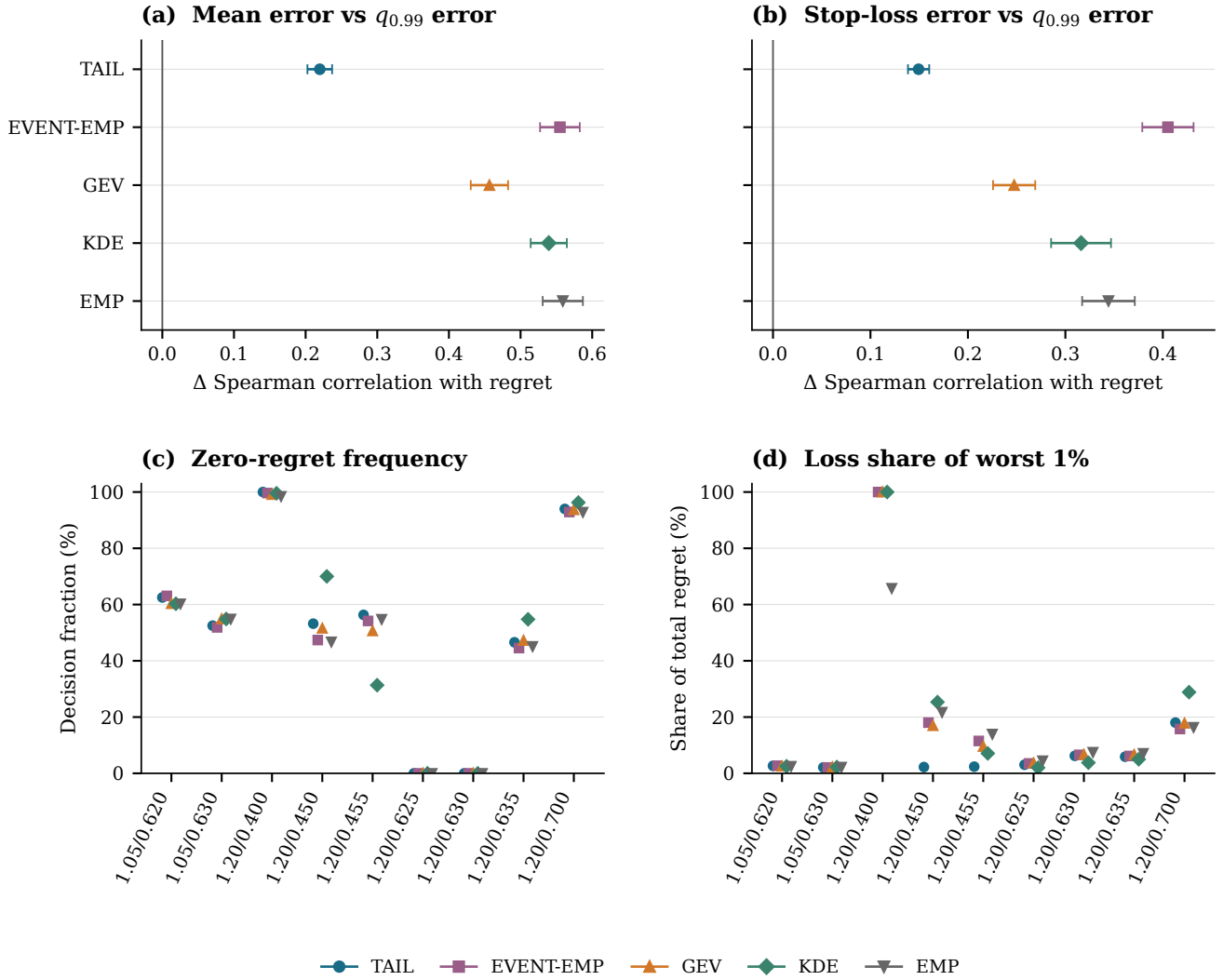


FIGURE S2 Functional-error associations and loss concentration. (a,b) Within-method differences between the Spearman correlation of regret with absolute mean or stop-loss relative error and its correlation with absolute $q_{0.99}$ relative error. Error and regret quantities are first averaged over nine cells within each of 10,000 histories. Intervals are the supplied simultaneous $\max_{|T|}$ 95% intervals for the family of ten correlation differences, based on 2,000 paired-history bootstrap resamples. They are not pointwise intervals. (c) Zero-regret frequency and (d) the share of aggregate regret accounted for by the worst 1% of decisions within each method–cell sample. The lower panels are descriptive summaries from 10,000 histories per method and cell. Offset, unconnected points distinguish methods at discrete tariff cells; the positions do not form a continuous tariff response.

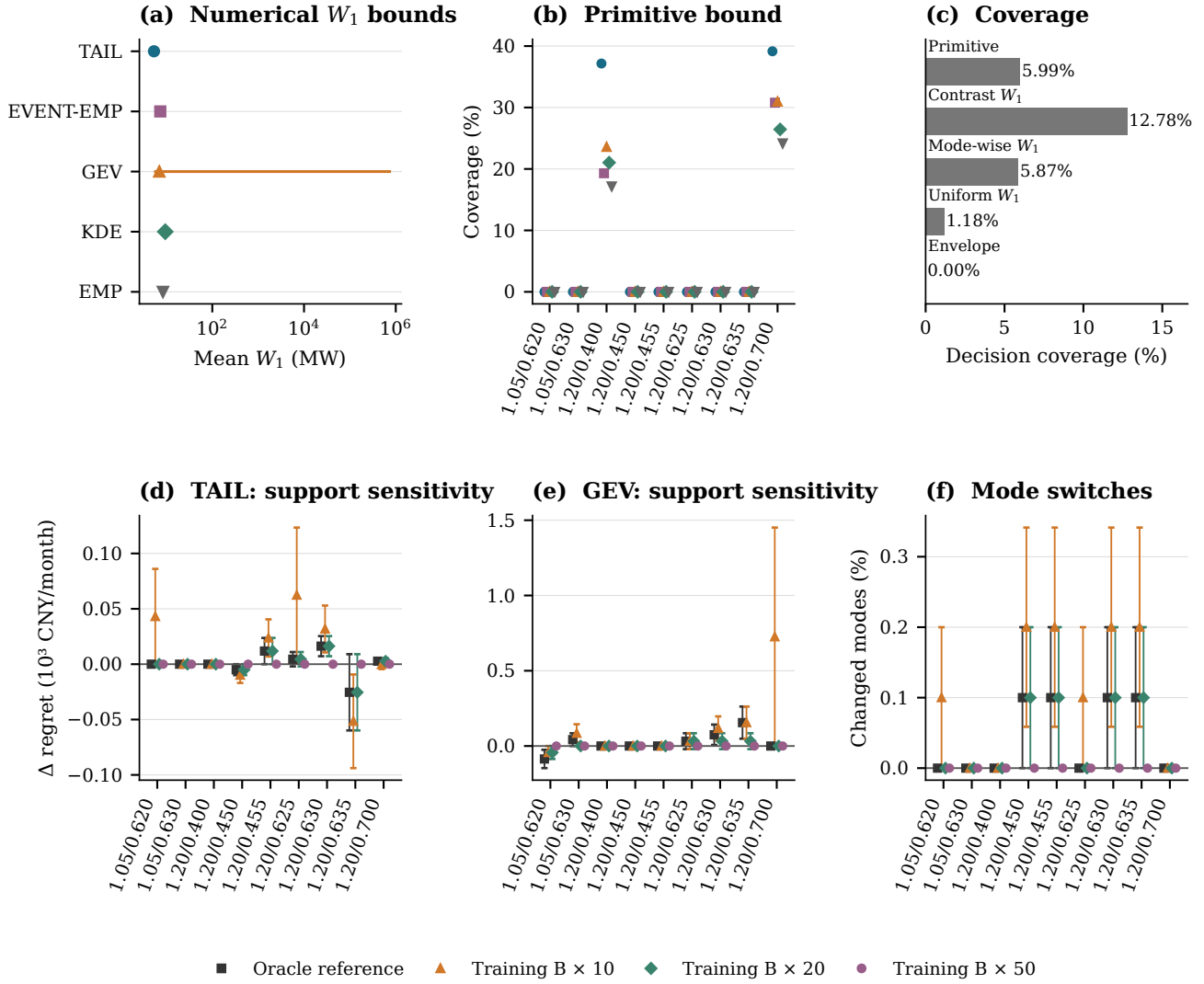


FIGURE S3 Numerical distribution-to-decision bounds and fitting-support sensitivity. (a) Mean W_1 point estimates and CDF-based numerical lower and upper bounds over 10,000 fitted laws per method; the logarithmic axis retains the loose GEV upper bound. (b) Cellwise coverage of the primitive numerical sufficient-condition check by predictive method. (c) Overall coverage of the Primitive, contrast-Wasserstein, mode-wise pairwise-Wasserstein, uniform-Wasserstein, and coarse distributional-envelope checks over 450,000 primary decisions. The first four use the same numerical full-action solver allowance. The contrast-Wasserstein and mode-wise pairwise-Wasserstein checks differ in whether the common distributional perturbation is bounded directly for each tariff contrast or after separate mode-value bounds. (d,e) Paired mean regret changes for TAIL and GEV under alternative fitted-support rules relative to the primary specification. (f) Percentage of TAIL tariff-mode decisions that change under each support rule. Error bars in (d–f) are one paired Monte Carlo standard error over the first 1,000 primary histories. Training-derived support rules use multipliers 10, 20, and 50; Oracle reference denotes the diagnostic bound $10q_{0.999}$. Point estimates in (a), the CDF-based bounds, and the solver allowances are numerical rather than interval-certified quantities.

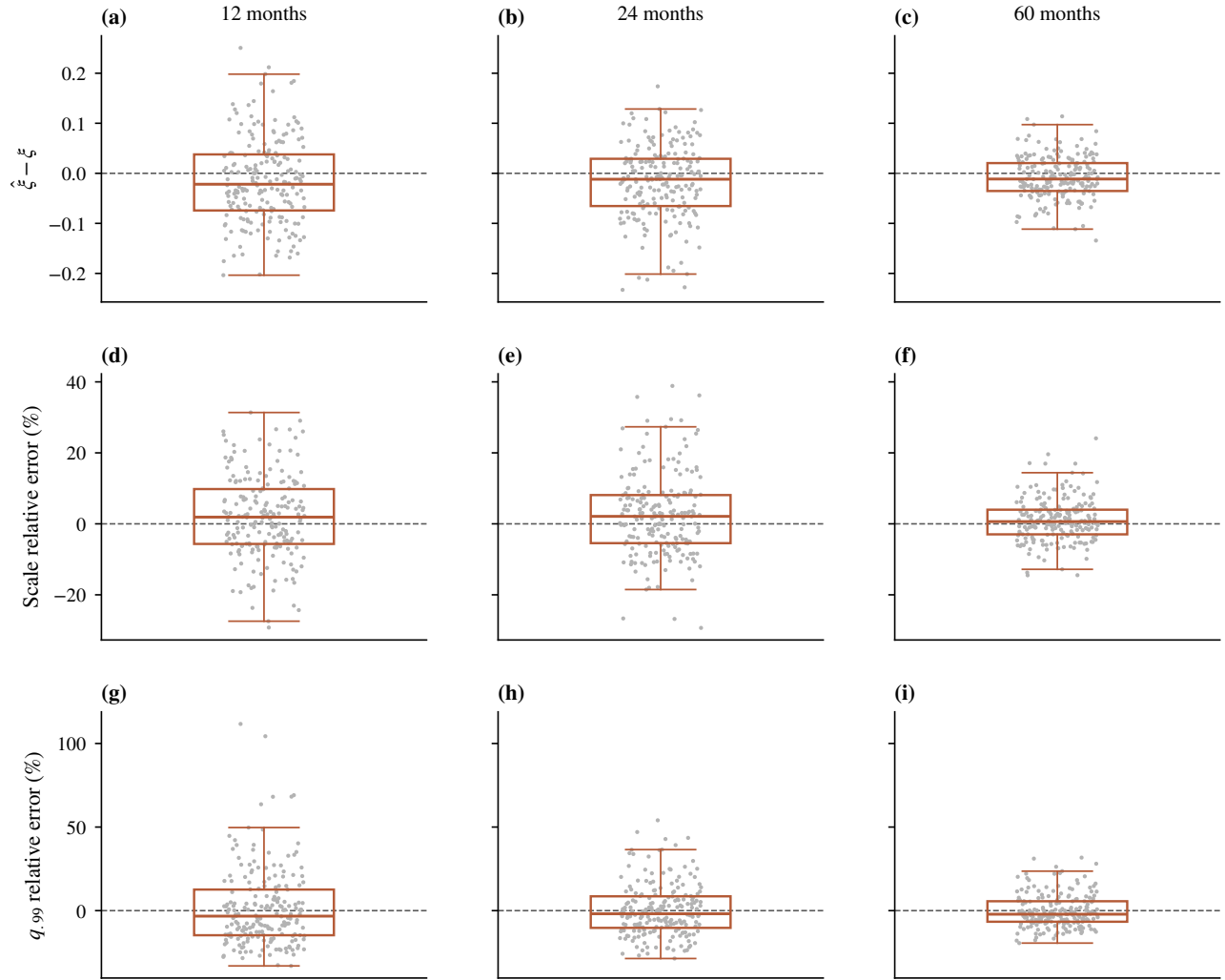


FIGURE S4 Parameter-recovery experiment. Columns show training lengths of 12, 24, and 60 months; rows show GPD shape error, relative scale error, and relative $q_{0.99}$ error. Each panel displays all 200 histories as grey points, with median, interquartile box, and 1.5-IQR whiskers. These are distribution summaries, not confidence intervals; the recovery experiment is separate from the primary analysis.

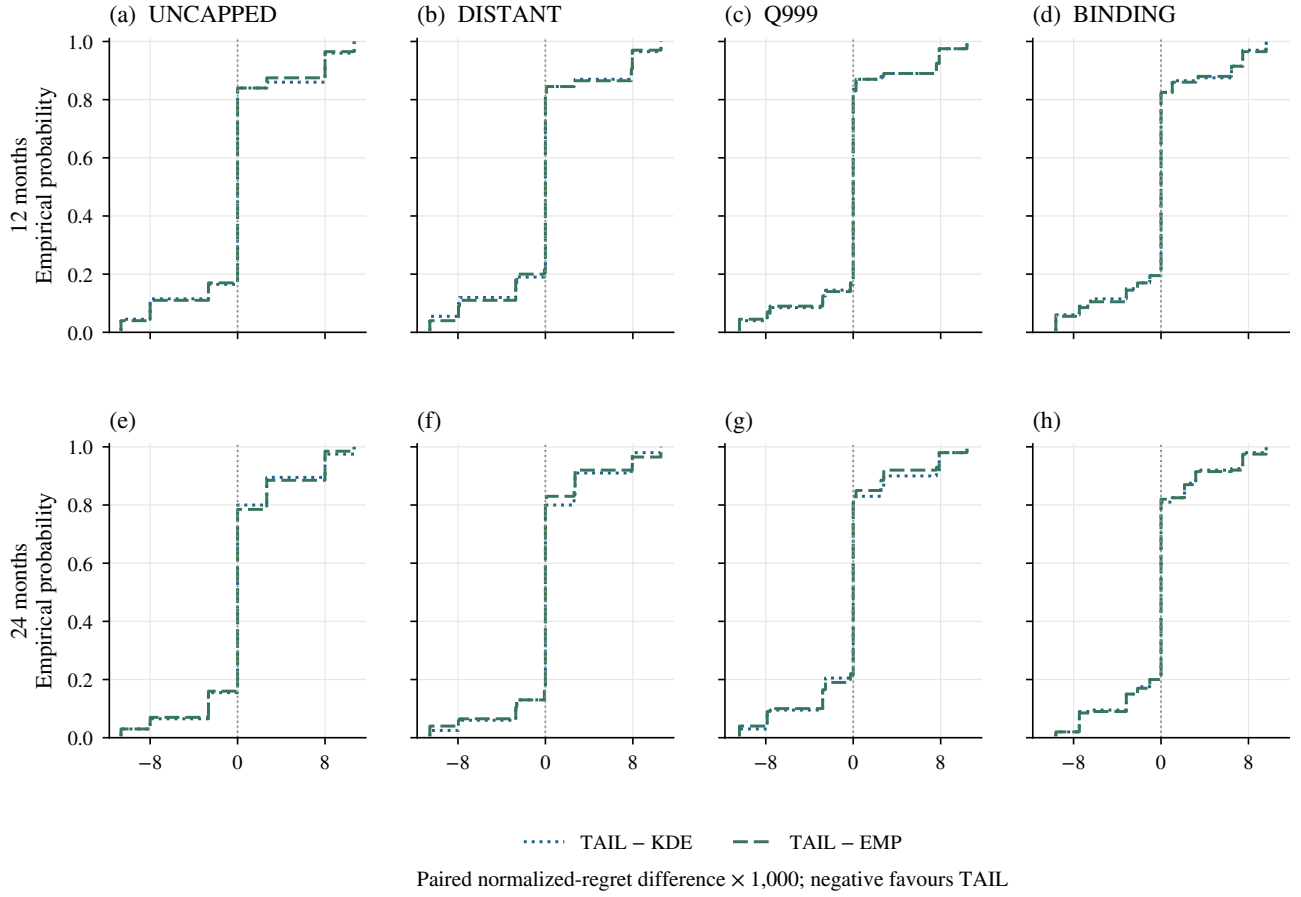
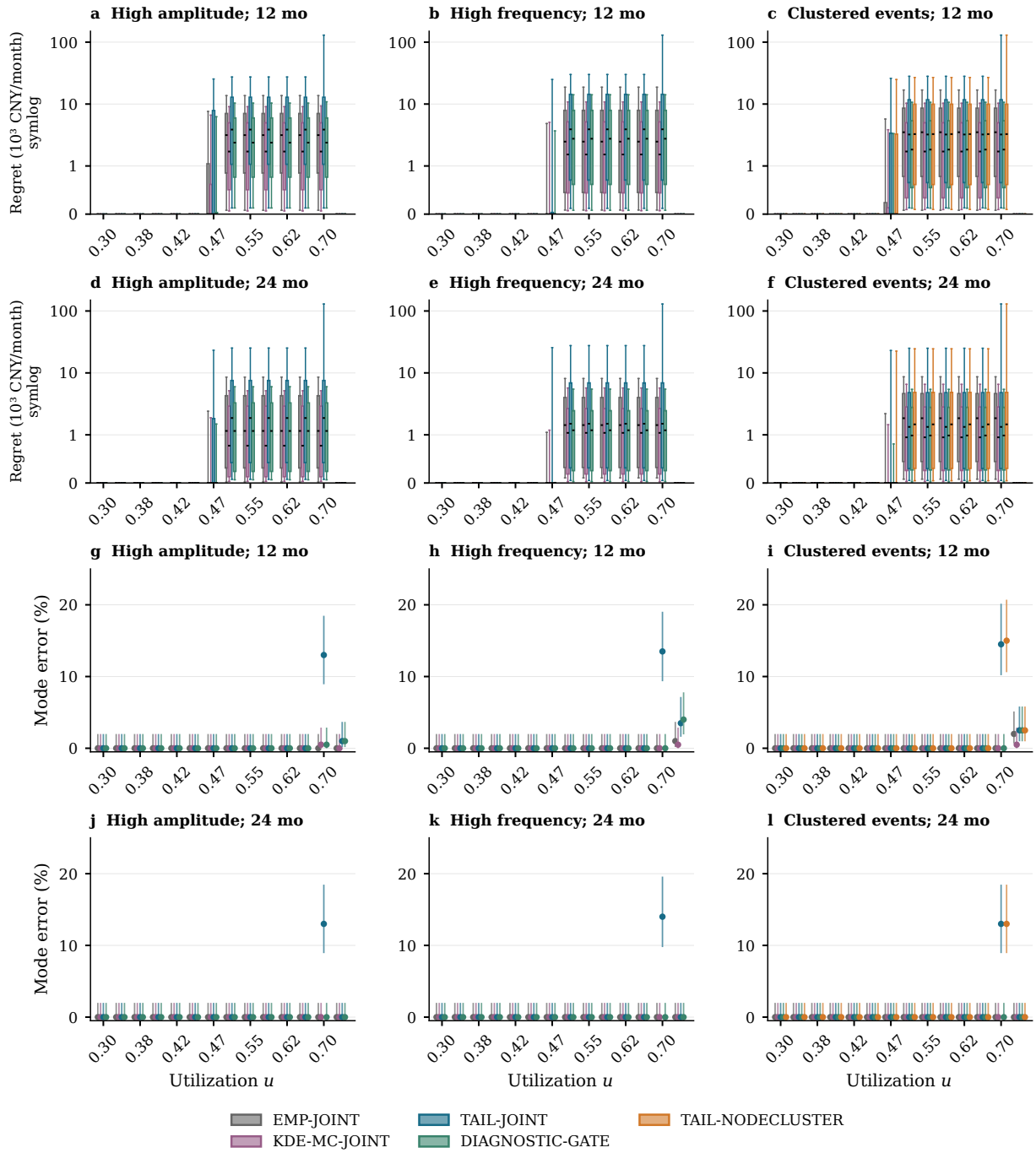


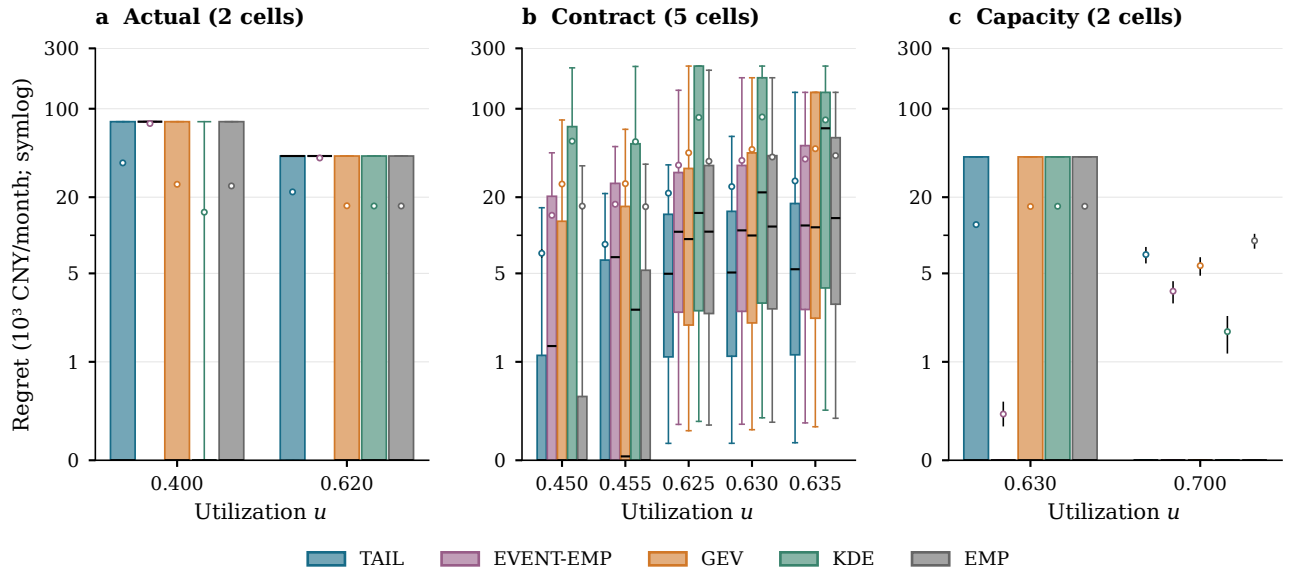
FIGURE S5 Physical clipping sensitivity. Columns show the four frozen clipping regimes and rows show 12 or 24 training months. Each ECDF contains 200 paired history differences in normalized regret times 1,000; negative values favour TAIL. The full data range is shown, and the interquartile range is zero in all 16 groups. This descriptive robustness experiment is separate from the primary analysis.



200 histories per cell; $\alpha = 1.20$. Boxes: P25–P75; whiskers: P10–P90; median tick.

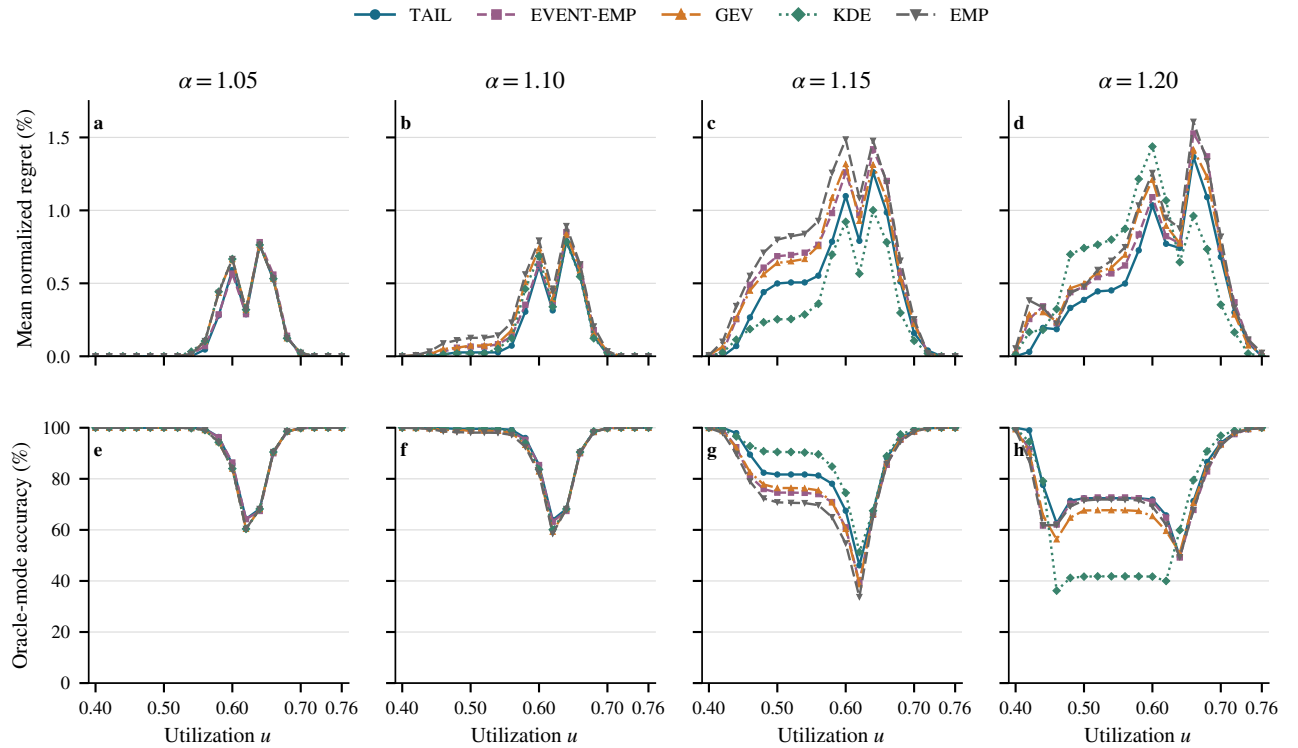
Lower panels: mode-error rates and Wilson 95% intervals; no interpolation between tariff cells.

FIGURE S6 Physical-pipeline experiments at $\alpha = 1.20$ using 200 paired histories per method, training length, and tariff cell (72,800 decisions). Columns represent high amplitude, high frequency, and clustered events. (a–f) Regret distributions at all 14 utilization cells for training lengths of 12 and 24 months: boxes span P25–P75, whiskers span P10–P90, and black ticks denote medians. The symlog ordinate retains zero and is linear below one thousand CNY per month. Outlying histories beyond the whiskers are not drawn individually; these are distribution summaries, not confidence intervals. (g–l) Mode-error proportions and pointwise Wilson 95% intervals at the same cells. Points are offset within each cell to separate methods and are not interpolated between cells. TAIL-NODECLUSTER applies only to the clustered-event scenario.



Boxes: P25–P75; black median; whiskers: P10–P90; open circle: mean. 10,000 histories per cell.

FIGURE S7 Cellwise regret distributions under the negative-binomial–Weibull challenge, with 10,000 histories per method and cell. Panels group the two actual, five contract, and two capacity oracle cells; labels give utilization and cells are equally spaced. Filled boxes span P25–P75, black ticks denote medians, whiskers span P10–P90, and open circles denote means. Black intervals at means use the formal pointwise normal 95% approximation, mean plus or minus 1.96 standard errors. The symlog ordinate is linear below one thousand CNY per month and logarithmic above it. Histories outside the displayed quantile ranges are not drawn individually. Quantile ranges describe history variability and are distinct from uncertainty in the mean.



Each method-cell profile uses 1,000 shared histories; lines connect the 19 evaluated utilization cells.

FIGURE S8 Complete five-method profiles over the final nonnegative, no-artificial-upper crossed tariff grid. Panels (a–d) show mean normalized full-action regret, R/V_F^* , and panels (e–h) show oracle-mode accuracy, for $\alpha = 1.05, 1.10, 1.15$, and 1.20 . Each panel covers the 19 evaluated utilization values; each method-cell estimate uses 1,000 shared histories, giving 380,000 decisions in total. Lines connect discrete evaluated cells for visual comparison and do not represent interpolation. The profiles are descriptive; no confidence intervals are shown and no invariant within-region ranking is asserted. TAIL and EVENT-EMP use event-level information; GEV, KDE, and EMP use monthly maxima.

TABLE S1 Same-information comparisons under high contract quantiles.

p^*	First – second	Oracle	Cells	Difference	95% CI
0.90	GEV – KDE	Actual	1	-12.19	[-13.02, -11.36]
0.90	GEV – KDE	Capacity	1	+12.97	[+12.26, +13.68]
0.90	GEV – KDE	Contract	8	-3.97	[-4.53, -3.41]
0.90	TAIL – EVENT-EMP	Actual	1	-8.80	[-9.73, -7.87]
0.90	TAIL – EVENT-EMP	Capacity	1	-20.68	[-22.18, -19.18]
0.90	TAIL – EVENT-EMP	Contract	8	-15.06	[-16.39, -13.72]
0.95	GEV – KDE	Actual	1	-18.40	[-19.46, -17.33]
0.95	GEV – KDE	Capacity	2	+3.28	[+1.49, +5.07]
0.95	GEV – KDE	Contract	7	+13.97	[+12.78, +15.15]
0.95	TAIL – EVENT-EMP	Actual	1	-24.59	[-25.86, -23.31]
0.95	TAIL – EVENT-EMP	Capacity	2	-113.74	[-117.77, -109.70]
0.95	TAIL – EVENT-EMP	Contract	7	-23.47	[-25.59, -21.35]
0.99	GEV – KDE	Actual	2	-52.28	[-54.06, -50.49]
0.99	GEV – KDE	Capacity	4	-318.31	[-329.72, -306.91]
0.99	GEV – KDE	Contract	4	-1.37	[-3.10, +0.36]
0.99	TAIL – EVENT-EMP	Actual	2	-80.28	[-82.37, -78.19]
0.99	TAIL – EVENT-EMP	Capacity	4	-1261.71	[-1288.14, -1235.28]
0.99	TAIL – EVENT-EMP	Contract	4	-70.45	[-73.18, -67.71]

Loss differences are in 10^3 CNY/month. Each comparison uses the same 10,000 fitted primary histories and equal weights over the indicated region cells; intervals are pointwise paired t intervals. Alpha is 2.20. No multiple-testing p-values are used for interpretation.

523 Supplementary tables

TABLE S2 Conditional within-contract regret by certificate status.

Split	Method	Certified	Decisions	Histories	Mean	95% CI
d_{MV}	TAIL	Yes	453	392	0.39	[0.34, 0.45]
d_{MV}	TAIL	No	17,142	6,812	7.81	[7.48, 8.15]
d_{MV}	EVENT-EMP	Yes	284	254	7.14	[5.86, 8.43]
d_{MV}	EVENT-EMP	No	17,581	6,948	11.73	[11.26, 12.20]
d_{MV}	GEV	Yes	228	208	4.23	[3.50, 4.95]
d_{MV}	GEV	No	16,308	6,658	13.11	[12.56, 13.65]
d_{MV}	KDE	Yes	144	131	12.19	[10.42, 13.96]
d_{MV}	KDE	No	10,684	4,273	9.75	[9.22, 10.28]
d_{MV}	EMP	Yes	221	201	8.80	[7.11, 10.48]
d_{MV}	EMP	No	17,285	6,995	17.86	[17.16, 18.56]
Pairwise	TAIL	Yes	1,081	684	0.67	[0.60, 0.73]
Pairwise	TAIL	No	16,514	6,601	8.02	[7.68, 8.36]
Pairwise	EVENT-EMP	Yes	714	465	7.61	[6.64, 8.58]
Pairwise	EVENT-EMP	No	17,151	6,813	11.73	[11.25, 12.21]
Pairwise	GEV	Yes	598	394	4.42	[3.81, 5.02]
Pairwise	GEV	No	15,938	6,526	13.24	[12.68, 13.79]
Pairwise	KDE	Yes	408	262	11.93	[10.71, 13.14]
Pairwise	KDE	No	10,420	4,220	9.58	[9.04, 10.11]
Pairwise	EMP	Yes	580	379	10.04	[8.47, 11.61]
Pairwise	EMP	No	16,926	6,891	17.84	[17.13, 18.55]

The subset contains decisions for which both oracle and fitted-law selected modes are contract. The mean is the equally weighted mean of eligible history means, with a pointwise t interval using the displayed number of histories. Loss units are 10^3 CNY/month. Different subsets overlap across methods and certificate splits; their counts must not be added to obtain the number of independent histories.

TABLE S3 Physical-mechanism stress tests at alpha 1.20.

Scenario	Months	Method	Regret	MCSE	Mode	Within	Error (%)
High amplitude	12	GATE	2.114	0.236	0.110	2.005	0.107
High amplitude	12	EMP-JOINT	2.512	0.248	0.000	2.512	0.000
High amplitude	12	KDE-MC-JOINT	1.783	0.200	0.046	1.737	0.036
High amplitude	12	TAIL-JOINT	5.740	0.653	1.267	4.473	1.000
High amplitude	24	GATE	0.997	0.090	0.000	0.997	0.000
High amplitude	24	EMP-JOINT	1.283	0.117	0.000	1.283	0.000
High amplitude	24	KDE-MC-JOINT	0.842	0.079	0.000	0.842	0.000
High amplitude	24	TAIL-JOINT	4.451	0.534	1.204	3.247	0.929
High frequency	12	GATE	2.589	0.277	0.253	2.336	0.286
High frequency	12	EMP-JOINT	2.731	0.274	0.063	2.668	0.071
High frequency	12	KDE-MC-JOINT	1.784	0.185	0.032	1.752	0.036
High frequency	12	TAIL-JOINT	5.937	0.621	1.472	4.464	1.214
High frequency	24	GATE	0.947	0.090	0.000	0.947	0.000
High frequency	24	EMP-JOINT	1.286	0.116	0.000	1.286	0.000
High frequency	24	KDE-MC-JOINT	0.948	0.095	0.000	0.948	0.000
High frequency	24	TAIL-JOINT	4.311	0.544	1.297	3.014	1.000
Clustered events	12	GATE	2.151	0.264	0.158	1.993	0.179
Clustered events	12	EMP-JOINT	3.135	0.315	0.127	3.009	0.143
Clustered events	12	KDE-MC-JOINT	1.855	0.217	0.032	1.823	0.036
Clustered events	12	TAIL-JOINT	5.303	0.582	1.502	3.801	1.214
Clustered events	12	TAIL-NODECLUSTER	5.380	0.595	1.548	3.832	1.250
Clustered events	24	GATE	0.936	0.093	0.000	0.936	0.000
Clustered events	24	EMP-JOINT	1.572	0.145	0.000	1.572	0.000
Clustered events	24	KDE-MC-JOINT	1.020	0.105	0.000	1.020	0.000
Clustered events	24	TAIL-JOINT	3.782	0.513	1.205	2.577	0.929
Clustered events	24	TAIL-NODECLUSTER	3.768	0.510	1.205	2.563	0.929

Each scenario-by-training-length summary uses 200 independent histories, each evaluated at 14 equally weighted tariff cells (2,800 decision rows). GATE denotes DIAGNOSTIC-GATE. Losses are in 10^3 CNY/month. MCSE is calculated after averaging cells within history. The oracle is a fixed independent synthetic test pool.

TABLE S4: Cell-specific NB–Weibull challenge results.

Region	Alpha / u	Method	Mean regret	95% CI	Accuracy (%)
actual	1.05 / 0.620	EMP	17.05	[16.64, 17.45]	59.7
capacity	1.05 / 0.630	EMP	16.96	[16.56, 17.36]	59.3
actual	1.20 / 0.400	EMP	24.55	[23.83, 25.27]	68.9
contract	1.20 / 0.450	EMP	17.01	[16.01, 18.01]	93.9
capacity	1.20 / 0.700	EMP	9.05	[7.83, 10.27]	97.9
contract	1.20 / 0.635	EMP	42.65	[41.53, 43.77]	81.3
contract	1.20 / 0.625	EMP	38.49	[37.23, 39.76]	90.1
contract	1.20 / 0.630	EMP	41.54	[40.31, 42.77]	86.2
contract	1.20 / 0.455	EMP	16.82	[15.86, 17.79]	94.8
actual	1.05 / 0.620	EVENT-EMP	40.83	[40.64, 41.02]	3.7
capacity	1.05 / 0.630	EVENT-EMP	0.47	[0.34, 0.60]	99.0
actual	1.20 / 0.400	EVENT-EMP	76.46	[76.18, 76.73]	3.2
contract	1.20 / 0.450	EVENT-EMP	14.39	[13.91, 14.87]	100.0
capacity	1.20 / 0.700	EVENT-EMP	3.62	[2.90, 4.34]	99.0
contract	1.20 / 0.635	EVENT-EMP	40.05	[39.01, 41.09]	77.4
contract	1.20 / 0.625	EVENT-EMP	35.85	[34.61, 37.08]	90.0
contract	1.20 / 0.630	EVENT-EMP	39.05	[37.86, 40.24]	84.5
contract	1.20 / 0.455	EVENT-EMP	17.60	[17.09, 18.10]	100.0
actual	1.05 / 0.620	GEV	17.11	[16.70, 17.52]	59.6
capacity	1.05 / 0.630	GEV	16.89	[16.49, 17.29]	59.4
actual	1.20 / 0.400	GEV	25.29	[24.56, 26.01]	68.0
contract	1.20 / 0.450	GEV	25.41	[24.24, 26.59]	91.2
capacity	1.20 / 0.700	GEV	5.75	[4.79, 6.71]	98.6
contract	1.20 / 0.635	GEV	48.33	[47.07, 49.58]	73.5
contract	1.20 / 0.625	GEV	44.86	[43.38, 46.34]	84.6
contract	1.20 / 0.630	GEV	47.69	[46.29, 49.10]	79.8
contract	1.20 / 0.455	GEV	25.59	[24.45, 26.74]	92.2
actual	1.05 / 0.620	KDE	17.05	[16.64, 17.45]	59.7
capacity	1.05 / 0.630	KDE	16.96	[16.56, 17.36]	59.3
actual	1.20 / 0.400	KDE	15.23	[14.62, 15.84]	80.7
contract	1.20 / 0.450	KDE	55.44	[53.71, 57.17]	75.9
capacity	1.20 / 0.700	KDE	1.73	[1.16, 2.30]	99.6
contract	1.20 / 0.635	KDE	81.62	[80.10, 83.14]	50.8
contract	1.20 / 0.625	KDE	85.43	[83.46, 87.40]	63.7
contract	1.20 / 0.630	KDE	86.07	[84.29, 87.84]	57.7
contract	1.20 / 0.455	KDE	54.95	[53.22, 56.67]	77.5
actual	1.05 / 0.620	TAIL	22.02	[21.61, 22.44]	47.9
capacity	1.05 / 0.630	TAIL	12.15	[11.78, 12.52]	70.8
actual	1.20 / 0.400	TAIL	37.34	[36.56, 38.11]	52.8
contract	1.20 / 0.450	TAIL	7.21	[6.66, 7.76]	98.3
capacity	1.20 / 0.700	TAIL	7.04	[5.99, 8.09]	98.3
contract	1.20 / 0.635	TAIL	26.88	[25.93, 27.83]	85.8
contract	1.20 / 0.625	TAIL	21.53	[20.53, 22.53]	94.1
contract	1.20 / 0.630	TAIL	24.26	[23.26, 25.26]	90.9
contract	1.20 / 0.455	TAIL	8.49	[7.96, 9.03]	98.6

524 10,000 histories per method and cell. Losses are in 10^3 CNY/month; intervals use mean ± 1.96 SE. The oracle design has two actual, five contract and two capacity cells.

TABLE S5: Complete regional paired NB–Weibull contrasts.

Region	First – second	Mean difference	95% CI
actual	TAIL – EVENT-EMP	-28.97	[-29.53, -28.40]
actual	TAIL – GEV	+8.48	[+7.98, +8.99]
actual	TAIL – KDE	+13.54	[+13.04, +14.05]
actual	TAIL – EMP	+8.88	[+8.37, +9.39]
actual	EVENT-EMP – GEV	+37.45	[+36.91, +37.99]
actual	EVENT-EMP – KDE	+42.51	[+42.03, +42.98]
actual	EVENT-EMP – EMP	+37.85	[+37.31, +38.38]

Region	First – second	Mean difference	95% CI
actual	GEV – KDE	+5.06	[+4.80, +5.32]
actual	GEV – EMP	+0.40	[+0.25, +0.55]
actual	KDE – EMP	-4.66	[-4.91, -4.41]
capacity	TAIL – EVENT-EMP	+7.55	[+7.10, +8.00]
capacity	TAIL – GEV	-1.73	[-2.19, -1.27]
capacity	TAIL – KDE	+0.25	[-0.28, +0.78]
capacity	TAIL – EMP	-3.41	[-3.91, -2.91]
capacity	EVENT-EMP – GEV	-9.28	[-9.70, -8.86]
capacity	EVENT-EMP – KDE	-7.30	[-7.74, -6.87]
capacity	EVENT-EMP – EMP	-10.96	[-11.49, -10.44]
capacity	GEV – KDE	+1.98	[+1.58, +2.37]
capacity	GEV – EMP	-1.68	[-2.04, -1.33]
capacity	KDE – EMP	-3.66	[-4.20, -3.12]
contract	TAIL – EVENT-EMP	-11.71	[-12.37, -11.05]
contract	TAIL – GEV	-20.70	[-21.69, -19.71]
contract	TAIL – KDE	-55.02	[-56.50, -53.55]
contract	TAIL – EMP	-13.63	[-14.49, -12.76]
contract	EVENT-EMP – GEV	-8.99	[-9.99, -7.99]
contract	EVENT-EMP – KDE	-43.31	[-44.77, -41.86]
contract	EVENT-EMP – EMP	-1.91	[-2.83, -1.00]
contract	GEV – KDE	-34.32	[-35.51, -33.14]
contract	GEV – EMP	+7.08	[+6.46, +7.69]
contract	KDE – EMP	+41.40	[+40.08, +42.71]

525 Within each history, cells in the oracle region are equally weighted. There are 10,000 paired history differences per contrast. Units: 10^3 CNY/month; intervals: mean ± 1.96 paired
526 SE. Negative differences mean lower regret for the first method.

TABLE S6: Numerical Wasserstein brackets.

Method	Mean W1 (MW)	Mean lower (MW)	Mean upper (MW)
EMP	8.39	8.14	8.63
EVENT-EMP	7.34	7.12	7.55
GEV	6.99	5.76	7.33e+05
KDE	9.43	8.92	9.94
TAIL	5.35	4.68	6.14

527 The W_1 point estimates are 256-point Gauss–Legendre numerical quadrature estimates. The CDF-based lower and upper columns are numerical bounds, not sampling confidence
528 intervals or interval-certified enclosures. For GEV, the point estimate and CDF calculations use the fitted law conditioned on its nonnegative natural support.

TABLE S7: Functional-error correlation contrasts.

Method	Contrast	Estimate	Simultaneous 95% CI
TAIL	Mean – q99	+0.220	[+0.203, +0.237]
TAIL	Stop-loss – q99	+0.149	[+0.139, +0.160]
EVENT-EMP	Mean – q99	+0.555	[+0.527, +0.583]
EVENT-EMP	Stop-loss – q99	+0.405	[+0.379, +0.432]
GEV	Mean – q99	+0.457	[+0.430, +0.483]
GEV	Stop-loss – q99	+0.247	[+0.226, +0.269]
KDE	Mean – q99	+0.539	[+0.514, +0.565]
KDE	Stop-loss – q99	+0.316	[+0.285, +0.347]
EMP	Mean – q99	+0.559	[+0.531, +0.587]
EMP	Stop-loss – q99	+0.344	[+0.317, +0.371]

529 The supplied max- $|T|$ intervals cover the family of ten within-method correlation differences. Resampling unit: history; 2,000 paired bootstrap resamples. Nine cells are averaged
530 within each of 10,000 histories.

TABLE S8: External fit-status counts, natural-tail variant.

Method	Formal fit status	Factory-origin fits
K-BLOCK	PASS	32
K-EMP	PASS	32
K-GEV	FALLBACK_EMPIRICAL	24
K-GEV	PASS	8
K-GTAIL	FALLBACK_EMPIRICAL	4
K-GTAIL	FALLBACK_EXPONENTIAL	13
K-GTAIL	PASS	15
K-KDE	PASS	32
K-POINT	PASS	32
K-UGPD	PASS	32

531 Eight factories and four rolling origins yield 32 fits per method. Labels are the formal stored status labels; fallback fits remain part of the evaluated pipeline.

TABLE S9: External predictive-score paired contrasts.

Score	Baseline	GTAIL – baseline	95% CI
crps	K-BLOCK	+5.627	[-9.534, +20.789]
crps	K-EMP	+3.005	[-13.858, +19.869]
crps	K-GEV	+3.002	[-14.318, +20.323]
crps	K-KDE	+4.654	[-11.430, +20.737]
qwrps_090	K-BLOCK	-0.510	[-1.895, +0.876]
qwrps_090	K-EMP	-0.510	[-1.895, +0.876]
qwrps_090	K-GEV	-0.613	[-2.177, +0.950]
qwrps_090	K-KDE	-0.087	[-0.938, +0.765]
q95_pinball	K-BLOCK	-2.594	[-10.152, +4.964]
q95_pinball	K-EMP	-2.594	[-10.152, +4.964]
q95_pinball	K-GEV	-3.268	[-11.774, +5.239]
q95_pinball	K-KDE	-0.548	[-5.256, +4.161]
q99_pinball	K-BLOCK	-4.071	[-11.427, +3.285]
q99_pinball	K-EMP	-4.071	[-11.427, +3.285]
q99_pinball	K-GEV	-4.481	[-12.648, +3.687]
q99_pinball	K-KDE	-0.050	[-2.356, +2.256]

532 Scores and intervals are multiplied by 10^3 . Each factory averages four origins; pointwise paired t intervals use eight factory differences and seven degrees of freedom. Score-specific
533 comparisons are descriptive.

TABLE S10: Conditional parameter-interval coverage

Months	Parameter	Covered / histories	Coverage (%) [95% MC CI]
12	xi	182 / 200	91.00 [86.15, 94.58]
12	sigma	190 / 200	95.00 [91.00, 97.58]
24	xi	185 / 200	92.50 [87.93, 95.74]
24	sigma	182 / 200	91.00 [86.15, 94.58]
60	xi	190 / 200	95.00 [91.00, 97.58]
60	sigma	191 / 200	95.50 [91.63, 97.92]

534 Separate inherited recovery experiment, 200 histories per setting; not the primary experiment. Coverage conditions on the selected threshold. Clopper–Pearson intervals describe
535 Monte Carlo uncertainty in coverage.

TABLE S11: Fitting-support policy sensitivity.

Alpha / u	Policy	TAIL delta R	SE	Mode (%)	GEV delta R	SE	Mode (%)
1.05/0.620	Oracle reference	+0.000	0.000	0.00	-0.086	0.061	0.20
1.05/0.630	Oracle reference	+0.000	0.000	0.00	+0.042	0.042	0.10
1.20/0.400	Oracle reference	+0.000	0.000	0.00	+0.000	0.000	0.00
1.20/0.450	Oracle reference	-0.005	0.005	0.10	-0.000	0.000	0.00
1.20/0.700	Oracle reference	+0.003	0.003	0.00	+0.000	0.000	0.00

Alpha / u	Policy	TAIL delta R	SE	Mode (%)	GEV delta R	SE	Mode (%)
1.20/0.635	Oracle reference	-0.025	0.034	0.10	+0.155	0.107	0.40
1.20/0.625	Oracle reference	+0.004	0.006	0.00	+0.032	0.053	0.20
1.20/0.630	Oracle reference	+0.016	0.009	0.10	+0.074	0.068	0.30
1.20/0.455	Oracle reference	+0.012	0.012	0.10	-0.000	0.000	0.00
1.05/0.620	Training x10	+0.043	0.043	0.10	-0.043	0.043	0.10
1.05/0.630	Training x10	+0.000	0.000	0.00	+0.085	0.060	0.20
1.20/0.400	Training x10	+0.000	0.000	0.00	+0.000	0.000	0.00
1.20/0.450	Training x10	-0.010	0.007	0.20	-0.000	0.000	0.00
1.20/0.700	Training x10	-0.000	0.004	0.00	+0.726	0.726	0.10
1.20/0.635	Training x10	-0.052	0.042	0.20	+0.155	0.107	0.40
1.20/0.625	Training x10	+0.062	0.061	0.10	+0.032	0.053	0.20
1.20/0.630	Training x10	+0.032	0.021	0.20	+0.117	0.080	0.40
1.20/0.455	Training x10	+0.024	0.017	0.20	-0.000	0.000	0.00
1.05/0.620	Training x20	+0.000	0.000	0.00	-0.043	0.043	0.10
1.05/0.630	Training x20	+0.000	0.000	0.00	+0.000	0.000	0.00
1.20/0.400	Training x20	+0.000	0.000	0.00	+0.000	0.000	0.00
1.20/0.450	Training x20	-0.005	0.005	0.10	-0.000	0.000	0.00
1.20/0.700	Training x20	+0.003	0.003	0.00	+0.000	0.000	0.00
1.20/0.635	Training x20	-0.025	0.034	0.10	+0.032	0.053	0.20
1.20/0.625	Training x20	+0.004	0.006	0.00	+0.032	0.053	0.20
1.20/0.630	Training x20	+0.016	0.009	0.10	+0.032	0.053	0.20
1.20/0.455	Training x20	+0.012	0.012	0.10	-0.000	0.000	0.00
1.05/0.620	Training x50	+0.000	0.000	0.00	+0.000	0.000	0.00
1.05/0.630	Training x50	+0.000	0.000	0.00	+0.000	0.000	0.00
1.20/0.400	Training x50	+0.000	0.000	0.00	+0.000	0.000	0.00
1.20/0.450	Training x50	+0.000	0.000	0.00	-0.000	0.000	0.00
1.20/0.700	Training x50	+0.000	0.000	0.00	+0.000	0.000	0.00
1.20/0.635	Training x50	+0.000	0.000	0.00	+0.000	0.000	0.00
1.20/0.625	Training x50	+0.000	0.000	0.00	+0.000	0.000	0.00
1.20/0.630	Training x50	+0.000	0.000	0.00	+0.000	0.000	0.00
1.20/0.455	Training x50	+0.000	0.000	0.00	-0.000	0.000	0.00

536 The TAIL and GEV column blocks each give paired mean change in regret, paired SE, and changed-mode percentage. All differences are relative to the primary support specification.

537 The first 1,000 paired histories form this separate support audit. Loss and SE units: 10^3 CNY/month. This subset does not replace the 10,000-history primary experiment.

TABLE S12 Complete history-paired regret contrasts in the primary experiment.

First – second	Actual	Contract	Capacity
	-0.61	-1.68	-2.67
TAIL – EVENT-EMP	[-0.77, -0.44]	[-1.93, -1.43]	[-3.24, -2.10]
	+0.82	-3.89	+5.47
GEV – KDE	[+0.68, +0.97]	[-4.14, -3.63]	[+4.87, +6.07]
	-1.66	+1.73	-8.71
KDE – EMP	[-1.88, -1.43]	[+1.40, +2.06]	[-9.41, -8.00]
	-0.66	-7.38	+4.13
TAIL – KDE	[-0.85, -0.47]	[-7.69, -7.06]	[+3.44, +4.81]
	-1.48	-3.49	-1.34
TAIL – GEV	[-1.72, -1.24]	[-3.77, -3.22]	[-1.98, -0.70]
	-2.32	-5.65	-4.58
TAIL – EMP	[-2.62, -2.01]	[-5.99, -5.31]	[-5.26, -3.89]
	-0.84	-2.16	-3.23
GEV – EMP	[-1.01, -0.66]	[-2.37, -1.95]	[-3.62, -2.85]
	-0.88	-1.81	+1.33
EVENT-EMP – GEV	[-1.12, -0.63]	[-2.05, -1.57]	[+0.77, +1.89]
	-0.05	-5.70	+6.80
EVENT-EMP – KDE	[-0.26, +0.15]	[-5.99, -5.40]	[+6.10, +7.50]
	-1.71	-3.97	-1.90
EVENT-EMP – EMP	[-1.99, -1.43]	[-4.25, -3.68]	[-2.46, -1.35]

First method minus second, in 10^3 CNY/month; negative values favour the first method. Brackets are pointwise paired t 95% intervals from 10,000 equally weighted regional history means (9,999 degrees of freedom). The first two comparisons share event histories and monthly maxima, respectively; the remaining comparisons provide descriptive context. Intervals are not simultaneous and are not interpreted as family-wise evidence.

TABLE S13 Primary tariff geometry and analytic demand-scale stability radii.

α	Utilization	Oracle mode	V_F^* (10^3 CNY/month)	γ_F (10^3 CNY/month)	Relative gap (%)	ρ_F^{contrast} (kW)	ρ_F^{pair} (kW)	ρ_F (kW)
1.05	0.620	Actual	5,336.473	43.036	0.806	1,195.45	1,195.45	298.86
1.05	0.630	Capacity	5,294.120	42.353	0.800	1,176.47	1,176.47	294.12
1.20	0.400	Actual	5,336.473	318.649	5.971	8,851.37	2,950.46	2,212.84
1.20	0.450	Actual	5,336.473	4.962	0.093	137.84	45.95	34.46
1.20	0.455	Contract	5,324.612	11.861	0.223	329.48	109.83	82.37
1.20	0.625	Contract	5,276.065	60.408	1.145	839.00	559.34	419.50
1.20	0.630	Contract	5,276.065	18.055	0.342	250.77	250.77	125.38
1.20	0.635	Capacity	5,252.434	23.631	0.450	328.20	328.20	164.10
1.20	0.700	Capacity	4,764.708	511.357	10.732	7,102.18	7,102.18	3,551.09

V_F^* and γ_F are in thousand CNY/month; relative gap is $100\gamma_F/V_F^*$. The direct contrast radius is $\rho_F^{\text{contrast}} = \min_{m \neq n_F^*} \gamma_{F,m}/K_{m,m_F^*}$; the mode-wise pairwise radius is $\rho_F^{\text{pair}} = \min_{m \neq n_F^*} \gamma_{F,m}/(L_m + L_{m_F^*})$; and the uniform radius is $\rho_F = \gamma_F/(2L)$ with $L = r_d \max\{1, \kappa\}$. All are oracle-dependent analytic quantities in kW and contain no numerical-integration or solver allowance. For the primary coefficients, $\kappa = 2$, so $K_{\text{act,con}} = 36$ CNY/kW compared with $L_{\text{act}} + L_{\text{con}} = 108$ CNY/kW. Across these cells, $\rho_F^{\text{contrast}} \geq \rho_F^{\text{pair}} \geq \rho_F$.

TABLE S14 Numerical sufficient-condition implementation audit.

Condition	Passing rows	Coverage (%)	Observed mode-selection violations
Primitive numerical check	26,975	5.994444	0
Contrast-Wasserstein numerical check	57,501	12.778000	0
Mode-wise pairwise-Wasserstein numerical check	26,393	5.865111	0
Uniform-Wasserstein numerical check	5,329	1.184222	0

The primitive check is $2\varepsilon_{\text{num}} + \zeta_{\text{full,num}} < \gamma_F$. The contrast-Wasserstein check requires $K_{m,m_F^*} W_{1,\text{upper}} + \zeta_{\text{full,num}} < \gamma_{F,m}$ for every competing mode, using the direct tariff-contrast constants defined in Section S1.7. The mode-wise pairwise check replaces K_{m,m_F^*} by $L_m + L_{m_F^*}$; the uniform check is $2LW_{1,\text{upper}} + \zeta_{\text{full,num}} < \gamma_F$, where $L = r_d \max\{1, \kappa\}$. All four use the same numerical full-action solver allowance. No observed mode-selection violations occurred among rows satisfying any of these checks. This is an implementation-audit result using floating-point numerical bounds, not an interval-certified numerical guarantee.

538 AI-ASSISTED PREPARATION

539 OpenAI Codex was used to assist with code development and checking, analysis implementation, manuscript revision, language
540 editing, and preparation of submission materials. The authors take responsibility for the content and interpretation of the work.

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